

Tracing Memories Over Time Using fMRI and Behavior

by

David J. Halpern

A DISSERTATION SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE DEGREE OF

DOCTOR OF PHILOSOPHY
DEPARTMENT OF PSYCHOLOGY
NEW YORK UNIVERSITY

SEPTEMBER, 2020

Todd M. Gureckis

© DAVID J. HALPERN
ALL RIGHTS RESERVED, 2020

Abstract

For forty years, cognitive neuroscientists have investigated the neural correlates of successful memory formation using neuroimaging. However, what exactly these signals represent and how they evolve with learning over time remain open questions. In this thesis, I leverage a large data set of 57 participants performing second-language learning task while in an MRI scanner to gain more insight into neural signals that accompany learning. It is commonly assumed that at least some signals that correlate with successful encoding reflect moment-to-moment fluctuations in internal states like attention and readiness to encode. In the first chapter, I test this assumption by asking whether both univariate and multivariate neural signals that have been shown to correlate with successful encoding contain information about whether an item will be remembered beyond what could be predicted based on that item's average memorability in the population. If they did reflect endogenous states of attention and readiness, they would be expected to improve prediction. However, I find that they do not, suggesting that these particular signals, in this task, may be representing features of the items being studied rather than variation in endogenous states. In order to identify novel signals that might track a participant's learning, the second chapter uses cognitive modeling and a large data set of 150 subjects performing the same task outside the scanner to make precise predictions about the evolution of memory signals over time. Using these techniques, I show that a model-based approach can identify memory-related signals that are not detected by standard approaches. These signals occur in learning and memory related regions such as the caudate nuclei suggesting that this model-based approach can be a more sensitive method for tracking learning over time with fMRI. Finally, the last chapter explores how combining large behavioral and fMRI data sets can improve inferences in general and the implications that this might have for the design of future cognitive neuroscience studies, particularly in studies of learning and memory where it is difficult to collect many trials in the same subject.

Contents

ABSTRACT	iii
LIST OF FIGURES	viii
DEDICATION	viii
ACKNOWLEDGMENTS	v
o INTRODUCTION	i
1 ITEM LEARNABILITY AND SUBSEQUENT MEMORY PREDICTION	2
1.1 Introduction	2
1.2 Methods	10
2 TRACING LEARNING OVER TIME: A MODEL-BASED SUBSEQUENT MEMORY ANALYSIS	33
2.1 Introduction	33
2.2 The OMNI Data set	38
2.3 Inferring knowledge states from behavioral and neural data	41
2.4 Results	57
2.5 Conclusions	60
3 GETTING BLOOD FROM A STONE: IMPROVING NEURAL INFERENCES WITHOUT ADDITIONAL NEURAL DATA	63
3.1 Neural Inferences	64
3.2 Two Sources of Error in Neuroimaging Analyses	69
3.3 Behavioral Data	75
3.4 Simulations	83
3.5 Collecting additional behavioral data	89
3.6 Conclusion	93

4	CONCLUSION	94
	APPENDIX A CLASSIFIER PLOTS	98
	REFERENCES	113

Listing of figures

1.1	Experimental Design.	12
1.2	Panel a shows the distribution of memorabilities (percent of subjects remembering the pair in a cued recall test) across all 45 word pairs. Panel b shows the distribution of subject abilities over all 57 subjects	23
1.3	Classifier performance based on the <i>Standard Subsequent Memory Model</i> . Each combination of ROIs, features and time-point treatment is plotted separately with definitions of terms found in <i>Methods</i> . <i>IPS</i> = Item Pattern Similarity, <i>GPS</i> = Global Pattern Similarity, <i>ISPC</i> = Intersubject Pattern Correlation	25
1.4	Classifier performance based on the <i>Full Model</i> . Each combination of ROIs, features and time-point treatment is plotted separately with definitions of terms found in <i>Methods</i> . <i>IPS</i> = Item Pattern Similarity, <i>GPS</i> = Global Pattern Similarity, <i>ISPC</i> = Intersubject Pattern Correlation. JOL and Baseline Behavioral Model are plotted in each column for comparison although they do not differ across columns.	26
1.5	Full Model results by study block. Each combination of ROIs, features and time-point treatment is plotted separately with definitions of terms found in <i>Methods</i> . <i>IPS</i> = Item Pattern Similarity, <i>GPS</i> = Global Pattern Similarity, <i>ISPC</i> = Intersubject Pattern Correlation. JOL and Baseline Behavioral Model are plotted in each column for comparison although they do not differ across columns.	27
2.2	The matrix of transition probabilities for either study or decay events in the two and three state model. The letters within each matrix reflect the transition parameters which are estimated to data. The state labels <i>U</i> are “unknown”, <i>K</i> are “known” (with possible forgetting), and <i>P</i> are “permanently known.”	45

2.3	Example illustration of the effect of fMRI observations on inferences about latent knowledge in a two state-model. A) Protocol showing the timing of study events for item 1 (black boxes) and item 2 (white boxes). B) State posterior estimates for item 1 obtained from a hypothetical setting of the two-state model parameters (dashed blue = S_U , solid orange = S_K). C) Hypothetical “observed” fMRI signal on each study trial for item 1 (inset shows the probability density function over MRI observation values conditioned on the state). D) State posteriors for item 1 after incorporating the observation likelihoods from study trials for this item. The inferred state probabilities are dramatically altered by the incorporation of the MRI observation (see text).	49
2.4	ROC curves for held-out predictions in each of the two-state (panel A) and three-state (panel B) model variants (<i>Recall</i> , <i>Recall+JOL</i> , <i>Recall+MRI</i>). The curves show the mean $\pm$ sem across each of the cross validation folds.	56
2.5	Examples of using estimated model to analyze the brain. A) Coronal slice showing left anterior hippocampal voxels tracking the change in S_K state posterior for each study trial. B) Topography (left; axial slice) and posterior predictive distributions (right) for MRI activations from most informative component in the three-state model. Individual traces show the distributions for each fold of the cross validation	58
3.1	25 samples from a multivariate Gaussian distribution are overlaid over an additional 100 samples from the same distribution. Regression lines are plotted showing how estimates from smaller samples can add significant variance to the estimated correlation. Error bars show that we cannot rule out a zero correlation in the smaller sample.	71
3.2	Here we show a case where the behavioral data is estimated with error. The correlation with the estimates is lower than it would be with the true behavioral data.	74
3.3	Anderson (1957) estimate ($\hat{\rho}$) of the correlation using all of the data as a function of Pearson’s r using only n_{nb} complete cases and the difference in the log of the variance of the behavioral data in the n_{nb} complete cases and all n_b cases	78
3.4	Correlation between the estimated $\hat{\theta}$ and θ from the hierarchical model, averaged over 10,000 simulated datasets, as a function of the number of neuroimaging subjects n_{nb} and number of additional behavioral subjects n_b .	86
3.5	Mean squared estimation error, averaged over 10,000 simulated datasets, for the estimate of ρ from the four correlation estimation methods as a function of the true correlation, the number of neuroimaging subjects n_{nb} and number of additional behavioral subjects n_b . We also compare to estimates from a neuroimaging dataset with $n_{nb} + 1$ subjects	86

3.6	Mean squared estimation error, averaged over 10,000 simulated datasets, for the estimate of ρ from the four correlation estimation methods as a function of the true correlation, the number of neuroimaging subjects n_nb and number of additional behavioral subjects n_b . We also compare to estimates from a neuroimaging dataset with $n_nb + 1$ subjects	87
3.7	Bias in estimate of ρ , over 10,000 simulated datasets, for the four estimation methods as a function of as a function of the true correlation, the number of neuroimaging subjects n_nb and number of additional behavioral subjects n_b . We also compare to estimates from a neuroimaging dataset with n_nb+1 subjects	89

TO MY PARENTS, GALE AND JOE, MY SIBLINGS, SARA AND DANIEL AND MY PARTNER,
MIREIA

Acknowledgments

Todo

0

Introduction

Todo

1

Item Learnability and Subsequent Memory Prediction

1.1 INTRODUCTION

When studying for a test in a language class, students might find that they easily recall some words but are not be able to retrieve others. What is it that makes a particular word more

likely to be remembered than another? Psychological studies of memory have uncovered many mechanisms by which an item might be more strongly encoded such as the spacing between multiple study sessions (Ruch, 1928), the order in which items are studied (Murdock, 1962) or the task used to encode the items (Craik and Tulving, 1975). In addition, items themselves have intrinsic, measurable differences in their average *memorability* in the population (Rubin, 1985; Rubin and Friendly, 1986; Nelson and Dunlosky, 1994; Grimaldi et al., 2010), at least some of which can be accounted for by features like frequency (DeLosh and McDaniel, 1996) or semantic centrality (Xie et al., 2020). However, there remains substantial variability in memory performance even after accounting for these factors (Kahana et al., 2018).

Cognitive neuroscience has opened up the possibility of measuring neural signals during memory encoding that can account for this additional variation in what gets remembered. The dominant approach to finding these signals has been through the subsequent memory paradigm (Paller and Wagner, 2002), where neural signals collected while encoding particular content are compared based on later memory performance. While such findings were first reported using ERPs (Sanquist et al., 1980; Paller et al., 1987), these were soon followed by fMRI studies, which had the ability to measure differences in neural activity with high spatial precision in healthy adults (Wagner, 1998; Davachi et al., 2003; Ranganath et al., 2003). Across these studies a coherent set of brain regions have been identified as being involved in human memory formation including the hippocampus and medial temporal lobe (Kim, 2011), which have long been associated with memory formation on the basis of animal and lesion studies (Scoville and Milner, 1957; Mishkin, 1978). In recent years, the field has moved beyond univariate activity, showing that the similarity of multi-

variate activity across presentations of the same item or other studied items is also related to memory recall and recognition (Rissman and Wagner, 2012; Xue, 2018) and suggesting particular computations that might be involved in memory formation.

Many researchers hope to interpret subsequent memory effects causally, measuring processes that make an item more or less likely to be remembered. For instance, at least some researchers claim that some subsequent memory analyses identify neural signals that index “moment-to-moment fluctuations of the quality of memory encoding” (Fukuda and Woodman, 2015) or activity “that supports successful encoding... [and] that leads to encoding failure” (Kim, 2011). However, a recent literature has suggested that the *memorability* of a particular item may drive a significant amount of brain activity during memory encoding (Bainbridge et al., 2017) and retrieval (Bainbridge and Rissman, 2018). Memorability poses a significant challenge to the causal interpretation of subsequent memory effects because it leads to a classic omitted variable bias: features that make an item memorable may lead to increased activity in brain regions that are unrelated to the memory encoding process per se. This activity would then be correlated with and predictive of subsequent memory but lack any causal link to the strength of encoding.

In addition, understanding subsequent memory effects requires decomposing these signals into those that are driven by external factors, such as those uncovered by psychological studies mentioned above, and those that can only be explained by endogenous factors (Weidemann and Kahana, 2019). In particular, the effects of endogenous shifts in attention or other variation in the quality or fidelity of a representation during encoding (perhaps due to variation in, e.g., personal semantic associations (Chadwick et al., 2016)) are often thought to underlie subsequent memory effects, especially multivariate ones (Xue, 2018).

Weidemann and Kahana (2019) previously found that subsequent memory effects based on oscillatory activity in scalp EEG were composed of a mix of both endogenous and exogenous factors. Deconfounding these effects in fMRI with known effects of serial position or item memorability is key to interpreting many results in the cognitive neuroscience of memory.

This deconfounding has the potential to affect fields outside of cognitive neuroscience as well. One of the original motivations (Sanquist et al., 1980) for searching for subsequent memory effects was that traditional cognitive psychology measurements like response times failed to capture the depth of encoding of an item (Craik and Tulving, 1975). Measures of subsequent memory have the potential to fulfill this role have already been used to make claims about cognitive mechanisms of learning such as the spacing effect (Feng et al., 2019). However, for a neural measure to provide a useful measurement of encoding processes, it would have to be able to predict individual variation in encoding after accounting for population-level effects on recall such as item and position. If we were to find such a signal, it could prove very useful for future cognitive modeling of learning and memory, particularly with modern joint modeling techniques (Turner et al., 2013, 2019). In addition, having independent measurements of learning are of great interest not only in the theoretical cognitive neuroscience and psychology literature but also in more applied fields and have the potential to inform educational interventions and automated tutoring systems (Atkinson, 1972; Corbett and Anderson, 1995).

1.1.1 APPROACH

In the following work, we attempt to understand the predictive power of fMRI signals of subsequent memory. We scan subjects while they are engaged in a Lithuanian-English second-language learning task where pairs of English words and their Lithuanian translation are presented one at a time. In order to identify signals that reflect endogenous processes, we attempt to predict subsequent cued recall in a held-out subject based on signals collected during encoding. We compare a baseline predictive model, that only includes information on the item being studied and subject studying it, to several models that also include various fMRI signals thought to relate to subsequent memory (Xue, 2018). Taking a similar approach to that of Weidemann and Kahana (2019) for EEG, we claim that if there are fMRI signals that improve classification performance over the baseline model, they likely reflect endogenous sources of encoding strength. Crucially, and unlike past work on subsequent memory effects including Weidemann and Kahana (2019), all subjects study items in the exact same order. At the expense of generalizability, this allows us to precisely quantify the effects of item memorability, presentation order and their interaction on subsequent recall and provides a stringent test for the fMRI signals.

SIGNALS

While early fMRI studies on subsequent memory effects focused on demonstrating differences in univariate activation (Kim, 2011; Paller and Wagner, 2002), subsequent work expanded the types of signals studied to include multivariate patterns (Rissman and Wagner, 2012; Xue, 2018). We therefore attempt to evaluate several of these signals in the same data set on their ability to predict memory on their own and after controlling for external

factors.

ITEM PATTERN SIMILARITY (IPS) The pattern similarity of an item with itself across repeated presentations has been shown to be correlated with memory in a number of studies, using both visual and verbal stimuli in recognition memory, associative fear learning, cued recall and free recall tasks (Xue et al., 2010, 2013; Ward et al., 2013; Visser et al., 2013; Bruett et al., 2020). Measuring item-to-item pattern similarity requires multiple presentations of the same item. Our study involves five repeated presentations so it presents an ideal opportunity to study these effects.

GLOBAL PATTERN SIMILARITY (GPS) In many regions, the similarity of a presentation of an item with all other studied items or all remembered items, has been shown to positively correlate with recognition memory with both verbal and visual stimuli (Bainbridge et al., 2017; Davis et al., 2014; LaRocque et al., 2013; Xiao et al., 2016). Global pattern similarity as a predictor of memory has often been theoretically motivated by models of recognition memory (Norman and O'Reilly, 2003; Gillund and Shiffrin, 1984; Nosofsky, 1988; Hintzman, 1988). However, there is good reason to think that it might be predictive of other kinds of memory as well. For instance, both hippocampal pattern separation and similarity across items is thought to increase the probability of retrieval in various memory tasks (Brunec et al., 2020). LaRocque et al. (2013) found that global pattern similarity in hippocampus was negatively correlated with recognition memory, which may be true across other tasks like cued and free recall. In addition, Kuhl et al. (2012) found that within-category similarity at encoding predicted later memory for category in a cued recall task. Because the categories in their task were broadly associated with brain regions (faces

and places), this predictive power could just be indexing proper processing of the stimuli. To the extent that the English and Lithuanian word pairs need to be processed both semantically and phonologically, we can consider all items in our study to be a part of a similar “category” or type of stimulus that requires similar processing.

INTER-SUBJECT PATTERN CORRELATION (ISPC) Several recent studies have explored the shared variance in neural signals while watching engaging stimuli like naturalistic videos (Chen et al., 2017). The across-subject temporal correlation of the same voxel (intersubject correlation, Hasson et al. (2008)) and the spatial correlation of the average pattern in an ROI (intersubject pattern correlation Koch et al. (2020)) while both subjects watch the same video are two signals that have been shown to correlate with later recollection of those videos. While our task did not use video stimuli subjects observed items in the task in the exact same order with the exact same timing. This suggests that it may be reasonable to treat our data in a similar way to data using video stimuli as responses to the stimuli should line up temporally.

As a comparison to the predictive models using the fMRI-derived features, we also ran a model using the Judgements of Learning as a predictor of subsequent memory. We use the raw numbers input by the subjects with no further preprocessing.

REGIONS OF INTEREST SETS

In order to test our ability to predict memory, we compare classifiers that each contain features from one of the feature sets described above (activity, item pattern similarity, global pattern similarity, inter-subject pattern correlation). All features require aggregating over a region and the same feature in several different regions may predict recall, possibly for

different reasons (Xue, 2018). Ideally we aim to strike a balance between including all potentially relevant signals and not including too many unrelated features. We therefore test classifiers using the following sets of regions described below.

TARGETED ROIs We first test a set of cortical ROIs derived from the existing cognitive neuroscience literature. We defined a targeted set of nineteen ROIs on the basis of prior expectations of these regions’ potential engagement in a learning task involving verbal materials. These ROIs included: left and right perirhinal cortex, left and right anterior temporal lobe, two separate areas in left anterior insula, left and right lateral occipital cortex, left and right dorsal parietal, left and right lateral parietal, left and right medial parietal (precuneus), left and right retrosplenial cortex, an upper and a lower portion of left ventrolateral prefrontal cortex (VLPFC), and visual word form area (VWFA; left hemisphere only).

HIPPOCAMPUS In a separate set of classifiers, we test features in a hippocampus ROI defined based on each participant’s individual anatomical scan. We test this separately both because of the central role the hippocampus is thought to play in memory formation (Scoville and Milner, 1957; Marr, 1971; Norman and O’Reilly, 2003) and because of the difficulty in recording fMRI BOLD signal from the hippocampus when using standard imaging sequences for targeting the whole brain as we do in this study (Litman and Davachi, 2008). The separate set means that the smaller signal-to-noise ratio in hippocampus BOLD will not prevent its inclusion in the penalized regression models we use for prediction (described in more detail below). We test classifiers that include the average activity, pattern similarity and global pattern similarity in each individual’s whole hippocampus

(*hipp-whole*) and also hippocampal subparts (left and right, posterior, medial and anterior, *hipp-sub*). In addition, we also select the voxels that are included in all participant’s anatomically-defined hippocampus which allows us to test a classifier based on individual hippocampal voxel activity (*hipp-vox*). Finally, we can define a whole hippocampus (*hipp-whole*) inter-subject pattern correlation feature based on these overlapping voxels.

SCHAEFER CORTICAL ROIs In addition to these two sets of theoretically motivated ROIs, we also include a set of 100 cortical ROIs defined based on resting-state functional connectivity (Schaefer et al., 2018). This allow us to be sensitive to any possible signals that might exist outside of the previous sets of ROIs.

WHOLE BRAIN Finally, one challenge in across-subject prediction is that voxels may not be perfectly aligned (Wang et al., 2020). Therefore, in order to be less sensitive to these differences (Rissman et al., 2010), we also test a set of classifiers where features are computed at the whole-brain level with each ”voxel” being the average activity in one of the 100 ROIs from Schaefer et al. (2018).

1.2 METHODS

1.2.1 BEHAVIORAL TASK

Starting with a normed set of Lithuanian-English words, we selected 45 translation pairs (Grimaldi et al., 2010). During study, participants saw the translation pairs presented one at a time for 4 seconds each with a variable duration inter-trial interval (4s-16s for consistency with event-related MRI timing). During this inter-trial interval, in order to prevent

rehearsal, participants made judgments about whether each of a series of numbers were odd or even. Words were presented on a computer screen with the Lithuanian word at the top of the screen and the English translation underneath.

Each word pair was presented five times and no pair was presented for the n th repetition until all words had $n - 1$ presentations. Importantly, and in contrast to many psychology studies on the subsequent memory effect, all participants see the same sequence of study items. At the expense of generalizability, this allowed us to precisely quantify the average recall performance for an word pair in a specific context and accounting for any interactions between items. The order of the word pairs was selected as follows: the 45 words were grouped into groups of five words each. On each repetition the order of the five words were shuffled but all five words appeared before the next group of five were presented. Immediately following the study session participants completed a second section outside the scanner that involved making judgments of learning (JOLs, (Nelson and Dunlosky, 1991)): for each pair participants were presented with the Lithuanian and English word and used the computer mouse to indicate on a scale of 0-100 how likely they were to remember the association in three days. Participants had up to twelve seconds to respond and the response was coded as missing if this deadline wasn't met (in practice all participants gave all JOL ratings within the allotted time).

Participants were then asked to return to the lab approximately 72 after the initial scan session for a recall test. Participants saw a Lithuanian word presented on the screen and had to type the associated English word. A trial was coded as correct if participants typed the correct English word (allowing for typographic errors) and all other responses were incorrect. Like the judgments of learning, this task was completed outside the scanner.

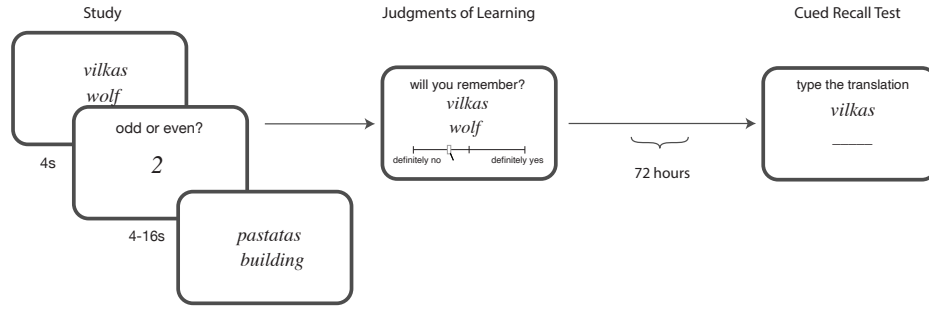


Figure 1.1: Experimental Design. On Day 1, in the fMRI scanner, subjects first completed a paired associates learning task in which an English word and a Lithuanian word were simultaneously presented on the screen. In between trials, subjects completed an odd-even judgment about a series of numbers in order to prevent rehearsal. After the learning session, subjects made a judgment of learning about each of the paired associates outside of the scanner. On Day 2, 72 hours after Day 1, subjects completed a cued recall test in which they were presented with a Lithuanian word and typed in the corresponding English word.

1.2.2 FMRI METHODS

MRI DATA ACQUISITION

MRI data were acquired on a 3 Tesla Siemens Prisma scanner.

Anatomical images were collected using a T₁-weighted MPRAGE high-resolution sequence (0.8mm isotropic voxels).

Five runs of whole brain BOLD functional data were acquired using an EPI sequence (2.5mm isotropic voxels; TR=1s; TE=35ms; Multiband factor=4; Phase Encoding: Anterior-Posterior).

An additional pair of non-accelerated EPI scans with opposing forward (anterior-posterior) and reverse (posterior-anterior) phase encoding relative to the primary functional data were collected for distortion correction during preprocessing (2.5mm isotropic voxels; TR=4.496; TE=45.6). The slice position and orientation for these scans matched those of the BOLD data collected during the behavioral task.

PREPROCESSING

Results included in this manuscript come from preprocessing performed using fMRIPrep 1.2.6-1 (Esteban et al., 2019), which is based on Nipype 1.1.7 (Gorgolewski et al., 2011).

ANATOMICAL DATA PREPROCESSING The T1w anatomical scans were corrected for intensity non-uniformity (INU) using N4BiasFieldCorrection (ANTs 2.2.0, Tustison et al., 2010). A T1w-reference map was computed (after INU-correction) using `mri_robust_template` (FreeSurfer 6.0.1, Reuter et al., 2010). The T1w-reference was then skull-stripped using `antsBrainExtraction.sh` (ANTs 2.2.0), using OASIS as target template. Brain surfaces were reconstructed using `recon-all` (FreeSurfer 6.0.1, Dale et al., 1999), and the brain mask estimated previously was refined with a custom variation of the method to reconcile ANTs-derived and FreeSurfer-derived segmentations of the cortical gray-matter of Mindboggle (Klein et al., 2017). Spatial normalization to the ICBM 152 Nonlinear Asymmetrical template version 2009c (Fonov et al., 2011) was performed through nonlinear registration with `antsRegistration` (ANTs 2.2.0, Avants et al., 2011), using brain-extracted versions of both T1w volume and template. Brain tissue segmentation of cerebrospinal fluid (CSF), white-matter (WM) and gray-matter (GM) was performed on the brain-extracted T1w using `fast` (FSL 5.0.9 Zhang et al., 2001).

FUNCTIONAL DATA PREPROCESSING For each of the five BOLD runs collected per subject (across all tasks and sessions), the following preprocessing was performed. First, a reference volume and its skull-stripped version were generated using a custom methodology of *fMRIPrep*. A deformation field to correct for susceptibility distortions was estimated

based on two echo-planar imaging (EPI) references with opposing phase-encoding directions, using 3dQwarp (AFNI 20160207, [Cox, 1997](#)). Based on the estimated susceptibility distortion, an unwarped BOLD reference was calculated for a more accurate co-registration with the anatomical reference.

The BOLD reference was then co-registered to the T1w reference using `bbregister` (FreeSurfer) which implements boundary-based registration ([Greve and Fischl, 2009](#)). Co-registration was configured with nine degrees of freedom to account for distortions remaining in the BOLD reference. Head-motion parameters with respect to the BOLD reference (transformation matrices, and six corresponding rotation and translation parameters) are estimated before any spatiotemporal filtering using `mcfliirt` (FSL 5.0.9, [Jenkinson et al., 2002](#)). The BOLD time-series (including slice-timing correction when applied) were resampled onto their original, native space by applying a single, composite transform to correct for head-motion and susceptibility distortions. These resampled BOLD time-series will be referred to as preprocessed BOLD in original space, or just preprocessed BOLD. The BOLD time-series were resampled to MNI152NLin2009cAsym standard space, generating a preprocessed BOLD run in MNI152NLin2009cAsym space.

First, a reference volume and its skull-stripped version were generated using a custom methodology of fMRIPrep. Several confounding time-series were calculated based on the preprocessed BOLD: framewise displacement (FD), DVARS and three region-wise global signals. FD and DVARS are calculated for each functional run, both using their implementations in Nipype (following the definitions by [Power et al., 2014](#)). The three global signals are extracted within the CSF, the WM, and the whole-brain masks. Additionally, a set of physiological regressors were extracted to allow for component-based noise cor-

rection (CompCor, Behzadi et al., 2007). Principal components are estimated after high-pass filtering the preprocessed BOLD time-series (using a discrete cosine filter with 128s cut-off) for the two CompCor variants: temporal (tCompCor) and anatomical (aCompCor). Six tCompCor components are then calculated from the top 5% variable voxels within a mask covering the subcortical regions. This subcortical mask is obtained by heavily eroding the brain mask, which ensures it does not include cortical GM regions. For aCompCor, six components are calculated within the intersection of the aforementioned mask and the union of CSF and WM masks calculated in T1w space, after their projection to the native space of each functional run (using the inverse BOLD-to-T1w transformation). The head-motion estimates calculated in the correction step were also placed within the corresponding confounds file. All resamplings can be performed with a single interpolation step by composing all the pertinent transformations (i.e. head-motion transform matrices, susceptibility distortion correction when available, and co-registrations to anatomical and template spaces). Gridded (volumetric) resamplings were performed using `antsApplyTransforms` (ANTs), configured with Lanczos interpolation to minimize the smoothing effects of other kernels (Lanczos, 1964). Non-gridded (surface) resamplings were performed using `mr_i_vol2surf` (FreeSurfer).

Preprocessed BOLD data were smoothed to 6mm FWHM using AFNI `3dBlurToFWHM` (https://afni.nimh.nih.gov/pub/dist/doc/program_help/3dBlurToFWHM.html).

REGIONS OF INTEREST (ROIs)

GRAY MATTER AND HIPPOCAMPUS ANATOMICAL MASKS Analysis of the functional data was restricted to voxels encompassed by a binary mask computed from the probabilis-

tic gray matter estimates generated during the Freesurfer anatomical preprocessing combined with anatomically defined voxels in individual hippocampi. For each participant, the probabilistic gray matter mask was binarized with threshold of 0.2 and intersected with bilateral hippocampal voxels as estimated using FSL FLIRT implemented in Nipype [Jenkinson and Smith \(2001\)](#); [Jenkinson et al. \(2002\)](#). The combined gray matter/hippocampus mask was resliced to functional resolution, smoothed with an 8mm FWHM kernel to be liberal in voxel inclusion (we opted to include some potentially non-gray matter voxels that could be ignored in downstream analyses rather than being overly strict and excluding relevant signal) and then re-binarized with a threshold of 0.2.

WHOLE BRAIN CORTICAL ROIs For whole brain cortical analysis, voxels were aggregated into ROIs provided by ([Schaefer et al., 2018](#)). To strike a balance between regional specificity and total number of cortical ROIs we used the intersection of the gray matter mask defined during preprocessing and the 100 parcel, 7 network atlas from [Schaefer et al. \(2018\)](#)* after reslicing to $2.5mm^3$.

TARGETED ROIs The set of targeted regions listed above regions were defined from a variety of sources. The perirhinal cortex (PRC) and anterior temporal lobe (ATL) ROIs were selected due to putative roles for these regions in processing conceptual information ([Martin et al., 2018](#)). The PRC regions were defined using the anterior portion (MNI $y > -21$) of the PRC ROIs provided by ([Ritchey et al., 2015](#)) as downloaded from Neurovault[†] which overlapped with the PRC center coordinates reported by [Martin et al.](#)

*https://github.com/ThomasYeoLab/CBIG/tree/master/stable_projects/brain_parcellation/Schaefer2018_LocalGlobal

[†]<https://neurovault.org/collections/3731/>; <https://identifiers.org/neurovault.collection:3731>

(2018). The ATL ROIs were defined as the Schaefer et al. (2018) ROIs encompassing the peak voxels reported by Martin et al. (2018) (Schaefer 600 parcellation ROIs 191 and 492).

The remaining targeted ROIs were defined by identifying voxel groupings from Schaefer et al. (2018) which encompassed other cortical regions expected to be involved in memory formation and/or processing of verbal materials involving concrete nouns (a group which includes all of the word pairs in our stimulus set) (Kim, 2011; Chen et al., 2019). The full set of targeted ROI definitions can be found at.

SINGLE TRIAL ACTIVATION ESTIMATES Individual voxel BOLD activation on each study trial was estimated with a general linear model (GLM) as implemented in SPM12 (Friston et al., 1994)[‡] using the *least squares-single* approach described in Mumford et al. (2012). For each study trial, a voxel-wise regression was carried out with one predictor containing a four second boxcar covering the timepoints corresponding to the to-be-estimated trial, another predictor containing four second boxcars covering the timepoints for all of the remaining trials, and fourteen additional predictors containing a variety of other potential confounds and nuisance signals as estimated during preprocessing (Power et al., 2014). Prior to estimation, the vectors of single-trial and all-other trials were convolved with the canonical SPM double gamma hemodynamic response function (HRF) along with its' temporal and dispersion derivatives (Lindquist et al., 2009). We then estimated regression coefficients for each voxel using ordinary least squares and the procedure was repeated for each study trial for each participant. The resulting coefficients for the single-trial predictors were taken as summaries for the activation for each voxel on each trial.

[‡]<https://www.fil.ion.ucl.ac.uk/spm/software/spm12/>

DECODING METHODS

After initial fMRI preprocessing, additional steps to create the fMRI features and reshape data were conducted using custom code in python 3.7.7 using the packages pandas (McKinney, 2010) and numpy (van der Walt et al., 2011) and R 4.0.2 using the package collections tidyverse (Wickham et al., 2019) and tidymodels (Kuhn and Wickham, 2020).

BASLINE BEHAVIORAL MODEL In order to control for item memorability, we first estimate a baseline model that accounts for word and presentation-order effects. To do so, we use a one-parameter IRT model.

$$P(r_{ws} = 1 | s, w) = \text{logit}^{-1}(\theta_s + b_w)$$
$$\theta_s \sim N(0, \sigma_\theta)$$

where w indexes Lithuanian-English word pairs, s indexes subjects, r_{ws} is the response (correct or incorrect recall) of each subject to each word pair) and b_w and θ_s represent the latent word learnability and subject ability parameters respectively. We estimate this model using maximum likelihood, as implemented in the R package lme4 (Bates et al., 2015).

FULL MODEL We use logistic regression as a model for predicting recall from the various proposed fMRI signals. Because our goal was to find features that improve predictions beyond our baseline model, we include a fixed intercept (or offset) in the full model which is the linear predictors for each subject and word pair that were estimated previously in the

baseline model, i.e. $(\hat{\theta}_s + \hat{b}_w)$. The full model we estimate is therefore:

$$P(r_{ws} = 1|s, w, \mathbf{X}) = \text{logit}^{-1}((\hat{\theta}_s + \hat{b}_w) + \beta \mathbf{x}_{ws}) \quad (1.1)$$

where $\mathbf{x}_{ws}$ is a vector of neural features (e.g. average activity in several ROIs) for subject s on word pair w .

STANDARD SUBSEQUENT MEMORY MODEL In addition to the full model, we also estimate a more typical subsequent memory model that simply predicts recall from the fMRI signals without controlling for the predictions from the baseline model. This model is as follows:

$$P(r_{ws} = 1|s, w, \mathbf{X}) = \text{logit}^{-1}(\alpha + \beta \mathbf{x}_{ws}) \quad (1.2)$$

In order to prevent overfitting and maximize predictive accuracy for both models including fMRI signals, we fit the model using maximum likelihood with an L_2 penalty on the β parameters, also known as ridge logistic regression (Hoerl and Kennard, 1970; Cessie and Houwelingen, 1992). This method has been used in several successful applications of MVPA techniques for studying the neural predictors of memory (e.g. Kuhl et al., 2012; Rissman et al., 2010; Weidemann and Kahana, 2019). To estimate the models, we use cyclical coordinate descent as implemented in the `glmnet` package (Friedman et al., 2010) in R. We estimate the predictive power of our models with leave-one-subject out cross-validation (Stone, 1974), using both the area under the ROC curve as metrics. We choose the size of the penalty on the β parameters (λ) using nested 10-fold cross-validation. We choose a set of λ hyperparameters to try using the default algorithm in `glmnet`: first $\lambda_{\max}$ is chosen such

that a model fit with that λ on all of the data will have all parameter estimates near 0, using a formula from [Friedman et al. \(2010\)](#). Then 100 parameters are chosen on a log scale from $\lambda_{\max}$ to $\frac{\lambda_{\max}}{10000}$. The model is estimated for each λ and the λ that maximizes the log likelihood of the held-out fold is chosen for evaluation. All of our fMRI-derived features are standardized (by their mean and variance in the training fold) before being included in the model. In addition, all of the correlation based features (*pattern similarity*, *global pattern similarity*, *intersubject pattern correlation*) were Fisher- z transformed ([Fisher, 1915](#)) prior to standardization.

To evaluate the models' generalization performance, we conduct statistical tests on the held-out AUCs. For the standard models, we compute a one-sample t-test, comparing the model's performance to the AUC of a random guessing model (.5). AUC is a useful metric here because it is not affected by the base rates of remembering and forgetting. For the main test, we compare the full models to the behavioral model using a paired t-test. To give the strongest possible chance to the MRI features, for all models, we use a *one*-sided test of whether it's AUC is greater. In leave-one(-subject)-out cross-validation, the distribution of AUC scores (or any other metric) across holdout sets will in general be correlated because the training set for the classifiers will be largely the same. The independence assumptions of the t-test are therefore violated ([Dietterich, 1998](#)). In order to remedy this, we use a permutation test to estimate the empirical null distribution of paired t -statistics when there is no relationship between the fMRI features and recall ([Pereira and Botvinick, 2011](#); [Pitman, 1937](#); [Stelzer et al., 2013](#); [Golland and Fischl, 2003](#)). To do so, we shuffle the pairing between the fMRI features and the Lithuanian-English word pair within subject. This allows us to maintain the correlation between fMRI features, as well as the distribution of

word-pair population memorabilities and subject recall abilities. Crucially, this also means that *Behavioral Baseline Model* does not change its predictions so the residual variance to be explained remains the same across permuted datasets. When shuffling the data, we also maintain the structure of the sequence of items such that the same five items appear in each group of five serial positions of each study block. We created 500 permuted datasets, applied to same classification models and computed a paired t -statistic for the permutation test.

One final complication is the fact that each item was shown multiple times in our study. Therefore, for each item, we have several observations that need to be combined to make a prediction. For the activity and inter-subject pattern correlation features there was one observation for each item in each of the five study blocks. For the pattern similarity features, there were observations for all possible pairs of study blocks. In the item pattern similarity feature, there were ten possible combinations of study blocks since the same item appeared in both of the patterns being correlated. In the global pattern similarity there were 20 possible combinations since each item in one of the five study blocks could be compared to items in the other four study blocks. To deal with combining these features, we tried three different approaches.

MEAN In our first approach, we simply took the mean of all of the observations as a single observation for that ROI and item. This approach has been used in previous studies with multiple presentations of the same item (e.g. [Xue et al., 2010](#)) and weighting a number of features equally can have good statistical properties over learning individual weights, especially in small datasets ([Dawes, 1979](#); [Schmidt, 1971](#)).

ALL In the second approach, we included all features on all repetitions, allowing the fitting and regularization to determine both which regions and which time points were the most relevant for prediction.

INDIVIDUAL BLOCKS/BLOCK PAIRS In the final approach, we fit each study block or pair of study blocks individually. This allows us for the possibility that the most relevant information comes only in earlier or later blocks. By not including as many potentially unrelated features as the second approach, the optimal amount of penalization is lower (Theobald, 1974). This means that the accuracy of the predictions will be higher. Within a feature and ROI set combination, we do a false discovery rate (FDR) adjustment to account for the 5, 10 or 20 tests that are done on each combination (Benjamini and Hochberg, 1995). Our tests are likely to be correlated within feature set across study blocks but this positive correlation is accounted for in the standard FDR procedure (Benjamini and Yekutieli, 2001). This makes these tests more comparable to the individual model tests using the other approaches.

1.2.3 RESULTS

BEHAVIORAL RESULTS

There was a large amount of variability in both subject memory abilities and word memorability scores in our data. Taking the average within each word pair, the median word pair was remembered by 37% of subjects with a standard deviation of 15% across all 45 word pairs. Taking the average within subjects, the median subject remembered 33% of the word pairs with a standard deviation of 26% across all 57 subjects. The distributions of word pair

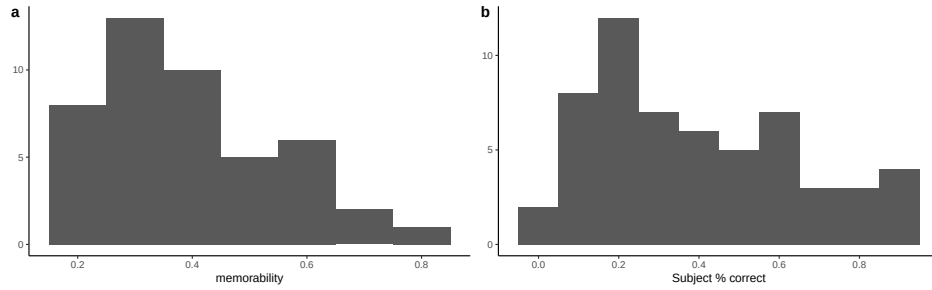


Figure 1.2: Panel **a** shows the distribution of memorabilities (percent of subjects remembering the pair in a cued recall test) across all 45 word pairs. Panel **b** shows the distribution of subject abilities over all 57 subjects

memorabilities and subject memory abilities are shown in figure 1.2

CLASSIFIER RESULTS

We first test whether our *Standard Subsequent Memory Model* can predict recall above chance using our proposed features and no other information. In figure 1.3, we plot the mean and standard error of the AUC estimates in held-out subjects. We show results using both the mean feature across all repetitions as well as using all study block and study block pairs. Several of our feature/ROI combinations, when aggregated at the mean level, predict memory significantly above chance (.5) based on a permutation test. These include all models including all Schaefer et al. (2018) ROIs, all models using *Targeted* ROIs except global pattern similarity (GPS) and GPS within the hippocampal subparts ($M = .538-.624$, $t_{56} = 2.59 - 7.75$, $ps < .01$). Results are overall similar when including each time point separately except that the Schaefer et al. (2018) intersubject pattern correlation (ISPC) and the hippocampal subparts model are no longer significant while the GPS model involving the whole hippocampus now is ($M = .527-.6$, $t_{56} = 1.83 - 6.5$, $ps < .05$). Because we are testing so many models it may make sense to control the overall false discovery rate across

all models tested. When we calculate the adjusted q values, the hippocampal models are no longer significant, along with the *Targeted* mean GPS model and the Schaefer et al. (2018) item pattern similarity (IPS) and ISPC models including all features.

Overall, this suggests that several features we tested are able to successfully predict memory across subjects and would be classified as subsequent memory effects in typical studies. In particular, when averaging over all study blocks and including all cortical ROIs, all of the features we derived from the literature could successfully predict memory in a heldout subject.

We now turn to the *Full Model* that controls for the memorability of individual items using the *Baseline Behavioral Model*. Figure 1.4 shows the results of the two models including the mean feature across all study-blocks and including all features. Figure 1.5 plots the results of using only a single study-block or study block pair for each classifier. We compare each model's held-out AUC performance to the baseline model using a paired t -test. None of our fMRI feature-based classifiers using the mean of the study-blocks or all study-blocks/study-block pairs perform significantly better than the *Baseline Behavioral Model* (all $p > .07$). In addition, after False Discovery Rate (FDR) adjustments (Benjamini and Hochberg, 1995), none of the classifiers using only a single feature predict significantly better than the *Baseline Behavioral Model* (all $q > .1$).[§]

In contrast to the classifiers with fMRI-derived features, the classifier that included judgments of learning did predict significantly better than the baseline model, improving

[§]Two of the classifier models, one including all hippocampal voxels from the first study block and one including the hippocampal inter-subject pattern correlation have undefined comparisons because they had minimal variation related to recall and the cross-validated penalty that was chosen meant that these features did not contribute to the prediction. Therefore, the difference between the AUCs in the baseline model and the two fMRI models was 0 and the variance was undefined, leading to an undefined t -statistic.

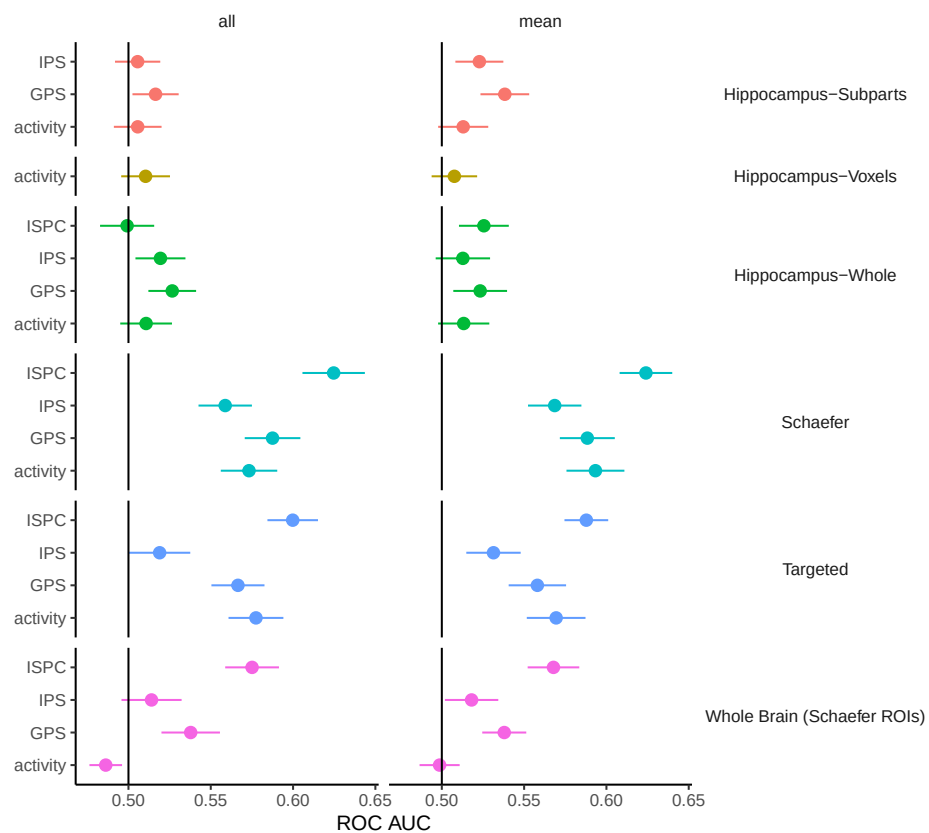


Figure 1.3: Classifier performance based on the *Standard Subsequent Memory Model*. Each combination of ROIs, features and time-point treatment is plotted separately with definitions of terms found in *Methods*. *IPS* = Item Pattern Similarity, *GPS* = Global Pattern Similarity, *ISPC* = Intersubject Pattern Correlation

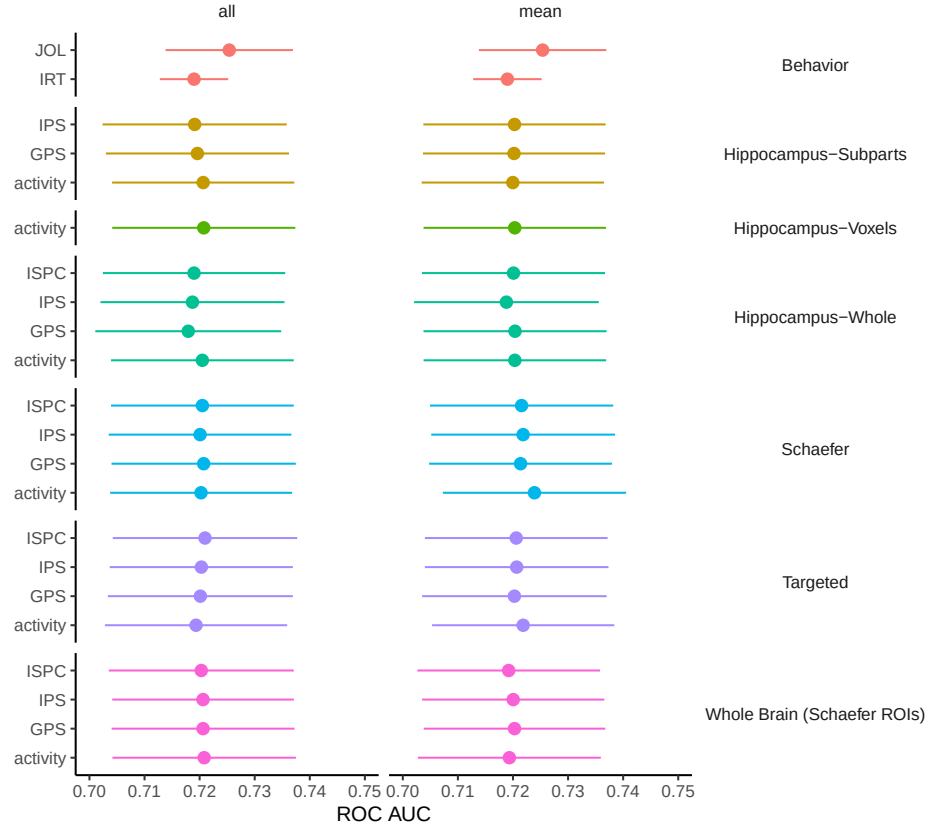


Figure 1.4: Classifier performance based on the *Full Model*. Each combination of ROIs, features and time-point treatment is plotted separately with definitions of terms found in *Methods*. *IPS* = Item Pattern Similarity, *GPS* = Global Pattern Similarity, *ISPC* = Intersubject Pattern Correlation. JOL and Baseline Behavioral Model are plotted in each column for comparison although they do not differ across columns.

the AUC on average by .0128 ($t_{(56)} = 6.84, p < .002$).[‡]

1.2.4 DISCUSSION

We found that several univariate and multivariate features of fMRI data proposed in the subsequent memory literature (Kim, 2011; Xue, 2018) could predict cued recall over a long

[‡]This was the smallest possible p-value in our permutation test given that we ran 500 samples. The true p -value may be much smaller but we do not have the precision to report it.

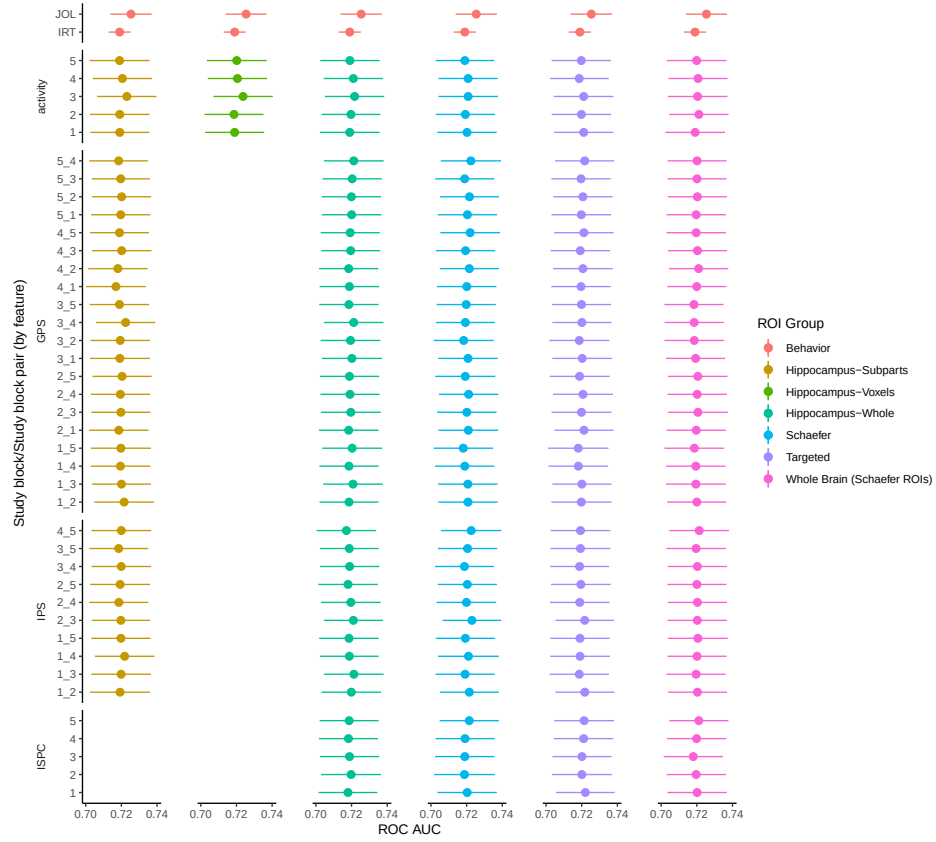


Figure 1.5: Full Model results by study block. Each combination of ROIs, features and time-point treatment is plotted separately with definitions of terms found in *Methods*. *IPS* = Item Pattern Similarity, *GPS* = Global Pattern Similarity, *ISPC* = Intersubject Pattern Correlation. JOL and Baseline Behavioral Model are plotted in each column for comparison although they do not differ across columns.

delay in a verbal paired associates task. To our knowledge, this is the first demonstration of subsequent memory effects in such a task. We also provide the first demonstration that Intersubject Pattern Correlation (ISPC) is a predictor of memory in a task not using video stimuli and that Global Pattern Similarity outside of hippocampus predicts memory in a recall task.

However, like many memory experiments, the items that we chose had a large amount of variability in their average memorability (Bainbridge, 2019) in the population. We therefore tested to what extent these effects could be explaining variation in recall that is not explained by population memorability. A common perspective on subsequent memory effects is that they index “the depth of encoding of the to-be-remembered stimuli” and “determine the efficacy of memory encoding” (Fukuda and Woodman, 2015). In contrast, our analyses, based on comparing predictive models using only behavior or behavior and fMRI data combined, showed that the subsequent memory effects we observed were highly correlated with population memorability and did not consistently improve predictions across subjects once the predictive model included the population memorability scores. This suggests that the interpretation of the source of subsequent memory effects may be more complicated than typically assumed. There has long been a recognition that “a specific experience may elicit the recruitment of the [processes indexed by subsequent memory effects] to a greater or lesser extent because of variable task demands, shifts in subjects’ strategies, characteristics of target items, or attentional modulations” (Wagner, 1998). At least in our data, these analyses suggest that objective characteristics of target items may drive a significant portion of the subsequent memory effect.

Of course, our analyses do not rule out the other sources of variability, particularly given

that, in our experimental design, we did not explicitly manipulate task demands, nor impose a strategy on subjects. In addition, the nature of our task (long delays, associative memory, novel stimuli), may have prevented some processes from being involved that may typically be measured by subsequent memory effects. However, the strength of the results using the *Standard Model* and the results of the *Full Model* including the judgments of learning place limits on these alternative explanations. In the following, we discuss several interpretations of these analyses and their implications for future studies of subsequent memory.

One explanation is that much of the variability in memory that could be predicted at encoding is explained by memorability already. For the fMRI features to predict when controlling for memorability, they would have to be reliably indexing endogenous processes that contradict the prediction from memorability, say a consistent failure of attention on certain trials for easy items. However, the performance of the model including Judgments of Learning shows that there is a signal (that is in fact consciously available to subjects) that consistently predicts subsequent memory even across items with the same population memorability. This model gives a lower bound on what is explainable at the time of encoding.

A second possible interpretation is that there is too much noise in the fMRI data to hope to find signals comparable to a judgment of learning in a data set our size. However, at 57 subjects, this dataset is approximately 2-3 times the size of most data sets in the subsequent memory literature, indicating that future work trying to isolate endogenous shifts in encoding processes may require large datasets and tightly-controlled designs.

A third possible interpretation is that there are indeed predictive signals that exist in

fMRI data but they are highly variable across people, perhaps due to a lack of precise voxel alignment or differently shaped brains causing variation in the signal-to-noise characteristics. If this were the case, it may be that our attempt to do across-subject prediction was doomed from the start. However, we point to the success of the *Standard Model* as evidence that these issues do not prevent any across-subject prediction in our data. It may be interesting to try to explain in future work why voxels that are correlated with memorability seem to be less sensitive to this type of variation. In addition, several of our features made efforts to circumvent this difficulty by a) aggregating data at higher levels (e.g. the whole brain analyses treating each ROI as a “voxel”), b) using hippocampal ROI definitions based on individual subjects’ anatomy and c) using multivariate measures like pattern similarity that are less sensitive to perfect alignment across subjects. However, none of these seem to have succeeded in improving classifier accuracy significantly.

A fourth possible interpretation is that the actual cognitive strategies used for learning (especially in an undirected, intentional encoding task like ours) highly variable across people. For instance, it may be that high memorability word pairs are easier to sound out, resulting in greater activity in phonological regions on average. But if sounding out is only an effective strategy for some people activity in a phonological region may predict memory but not consistently beyond the memorability score. Across-subject prediction differs from the usual two-level within-subject paradigm that is typical of MVPA studies in that it is more sensitive to small magnitude, low variability effects but less sensitive to high magnitude, high variability effects (Wang et al., 2020). If the measurement of encoding processes is more like the latter scenario, this suggests (like above) that future studies of subsequent memory will require larger data sets that allow for understanding groups of subjects who

use similar encoding strategies.

Overall, these four explanations can be summed up by a low signal-to-noise ratio and/or high signal but high across-subject variability in both the fMRI BOLD measurement as well as cognitive processes at the time of encoding (as a predictor of subsequent memory). We argue that the existence of the JOLs as an orthogonal predictor to memorability and the existence of successful memory prediction from fMRI in the absence of memorability information suggest that these can't be the complete story.

One additional possibility is that endogenous processes indexed by subsequent memory signals might not be simply randomly fluctuating but may actually fluctuate strategically in a way that is related to memorability. For instance, a subject might decide whether or not to allocate attention to a particular item based on whether they think it is possible to learn the item. We might hope that there would be some variation in this perceived memorability that is not captured by the population memorability but could be captured by fMRI signals. For instance, two words in a paired associates task might not be similar according to most people but might have been more closely paired based on a person's individual experiences (Charest et al., 2014). However, it is possible that controlling for memorability removes too much of this variance for it to be detectable in our data.

In the forty years since the first subsequent memory effect was reported, much has been learned about the neural signals associated with memory. A vast literature documents the various regions and multivariate signals, along with proposed cognitive computations being implemented and supporting evidence from animal and lesion studies. This paper contributes to this literature by documenting, for the first time, several aspects of subsequent memory effects in a novel language-learning task. We show that several features proposed in

for other types of memory tasks such as intersubject pattern correlation and global pattern similarity also apply here. However, the exact interpretation of these subsequent memory effects remains ambiguous and the original hope of identifying an item-level measure of depth-of-processing has still not been fully realized. Here we attempted to precisely control for stimulus and presentation-order effects that have been documented to have influence over the probability of recall in order to see whether we can extract signals from fMRI that correlate with moment-to-moment fluctuations in ability to encode. While we were not able to identify predictive signals, we suggest that future work leverage the techniques of this paper and [Weidemann and Kahana \(2019\)](#), novel classifier analyses and tight control of memorability, with larger datasets, different tasks and new features of the fMRI signal. In several ways, our own design could be improved, for instance by using items with more similar memorability scores (so that more variance can be explained by the neural signals), by including movie prior to the experiment that allow for the use of better alignment tools ([Haxby et al., 2011, 2020](#)).

2

Tracing Learning Over Time: A Model-Based Subsequent Memory Analysis

2.1 INTRODUCTION

A shared goal for both cognitive science and educational data mining is the development of accurate models of human learning. On the basic science side, learning and memory are

important functions of the human brain that support our ability to flexibly interact with our environment. On the education side, predictive theories of learning may be leveraged by intelligent tutoring systems (ITS) to individually optimize instruction (Anderson et al., 1995; Koedinger et al., 2013).

Perhaps the most influential approach to modeling student learning in the educational data mining literature is “knowledge tracing” (Atkinson, 1972; Corbett and Anderson, 1995) whereby the learned mastery of a particular skill or fact is treated as a latent state and the probability that a person’s knowledge is in that state is updated in light of observed student behavior. For example, in Bayesian Knowledge Tracing (BKT), each learning unit is assumed to be in one of two discrete states: $\{unknown, known\}$. Each time the student engages in a learning activity, the latent knowledge can transition from the *unknown* to the *known* state with probability l . Performance on a test, quiz, or exercise is conditional on the latent knowledge state, such that being in the *known* state is typically associated with a higher probability of issuing a correct answer than being in the *unknown* state. Using the model, it is possible to infer posterior probabilities of the knowledge state of each learner and skill using Bayes’ rule, given the pattern of responses made on various assessments or quizzes. These probabilities are then used to make predictions about learning performance for new students, as well as to design optimized instruction policies.

Research in this area focuses on building more precise models of student learning by, for instance, incorporating factors that reflect individual abilities (Yudelson et al., 2013; Liu and Koedinger, 2017), contextual factors that contribute to errors (Baker et al., 2008), or models of the exact moment at which a skill is acquired (Baker et al., 2011). However, one relatively underexplored question is what types of observable data may be most useful for

informing inferences about latent knowledge states during learning. Of particular interest is the idea that many other features besides overt responses might be partially informative. For example, the student's response time to a test question may add additional information about learning alongside correctness (Pavlik and Anderson, 2008; van der Linden, 2009; Wang and Heffernan, 2012). Likewise, patterns of mouse or eye movements during a learning session might help index drifting attention (Bondareva et al., 2013; Rau and Pardos, 2016).

In this paper we demonstrate that it is possible to integrate indirect neural measurements of brain activity into a cognitive model of learning in a way that 1) can improve prediction of a learner's test performance at a 72 hour delay and 2) allows knowledge tracing without interrupting the learning environment with explicit tests or assessments (which can be distracting or may bias learning).

Although acquiring neural recordings is impractical in most educational settings, the approach of fusing multiple sources of sensor data about individual learners may be a generally useful method for the educational data mining literature. In addition, as we show in our results, such modeling efforts may also feedback to contribute to a better understanding of the neural and cognitive mechanisms that support learning and memory (Anderson et al., 2010; Anderson, 2012; Turner et al., 2013, 2016). Finally, as the cost and difficulty of making indirect neural recordings falls (e.g., due to the advent of portable, dry contact electroencephalogram or EEG) the practicality of utilizing such sensors will likely increase (c.f., (Dikker et al., 2017)).

We begin by reviewing past work in cognitive neuroscience which has attempted to identify predictive signals of learning and memory processes. Next we describe our approach

fusing concepts from knowledge tracing with what is known about the cognitive neuroscience of memory. We then describe a dataset collected from human participants performing a simple second-language learning task while undergoing functional magnetic resonance imaging (fMRI). We compare the predictive power of a variety of models against held-out memory recall data at study-test delays ranging from one day to one week. From the fitted model we then extract the neural signals corresponding to learning in the study period.

2.1.1 PRIOR WORK USING COGNITIVE NEUROSCIENCE METHODS TO PREDICT INDIVIDUAL LEARNING

The prediction and optimization of human learning has been a long standing goal of cognitive neuroscience research. On the prediction side, a number of studies have explored the “subsequent memory” paradigm (Sanquist et al., 1980; Paller et al., 1987; Davachi, 2006; Ranganath et al., 2003). In these experiments, participants study controlled stimuli such as lists of word pairs while brain signals (such as the blood oxygen-level dependent “BOLD” signal measured via fMRI or event-related potentials, ERPs, assessed with EEG) are recorded. Some time later, participants’ memory is tested for the material they saw during study. Accuracy on each memory test item is used to back-sort the neural data recordings into brain patterns associated with successful versus unsuccessful later memory. Regions with a reliable difference in brain activation between these two classes are taken to reflect neural correlates supporting lasting memory formation. These studies have isolated specific regions including the hippocampus and medial temporal lobe as being associated with memory formation, aligning with previous findings from animal and lesion

studies (Scoville and Milner, 1957; Brown and Behrmann, 2017).

Building on this work, Fukuda and Woodman (2015) identified two EEG-based subsequent memory signals and used these to classify study trials in a memory experiment as likely to be remembered (*initially well studied*) or forgotten (*initially poorly studied*). In a subsequent session, participants were allowed to restudy half of the items identified as *initially well studied* and half of the items identified as *initially poorly studied*. A final test then assessed knowledge for all of the items. Of particular interest was the finding that the restudy opportunity most benefitted the *initially poorly studied* items compared to the other items. Importantly, the entire prediction about what was or wasn't well studied was based exclusively on indirect neural recordings for each subject rather than any explicit assessment or test.

The subsequent memory paradigm has been a powerful tool for studying the neural basis of memory. However, the cognitive neuroscience literature does not currently take advantage of the wealth of knowledge about predicting individual learning from the educational data mining and cognitive modeling literatures. For example, classifying brain patterns as forgotten based on a single test fails to account for the possibility of “slippage” (errors in performance of a mastered skill due to chance) which is central to BKT models (Corbett and Anderson, 1995). Likewise, when an item is not remembered at test it could be for a number of reasons: the item may have been poorly encoded during the study session, or perhaps was well encoded and would have been remembered at an earlier study session but was simply forgotten due to decay or interference. Structured models such as Hidden Markov Models (HMMs) can account for such latent memory dynamics and use them to help improve predictions. The subsequent memory approach is also difficult to

apply when learners get repeated study opportunities because of ambiguity about which brain scans should be classified as causally related to the test performance. Finally, the standards for model development within the machine learning and data mining communities is predictive performance on held-out data which is often more difficult than describing statistically reliable patterns within a single data set due to the ability to overfit.

To address these issues, we describe an approach to the simultaneous modeling of behavior and neural recordings in a single knowledge tracing model*. Our aim is to demonstrate the value of combining insights from these still somewhat disparate literatures. The approach we take is in some ways similar to work by Anderson and colleagues that has tried to infer from fMRI the mental state of individuals as they engage in complex math problems (Anderson et al., 2010; Anderson, 2012; Anderson et al., 2012; Zhang et al., 2016; Tenison et al., 2016) (see also Turner et al., 2013, 2016). While these reports hint at the utility of combining fMRI with probabilistic cognitive models, this prior work does not specifically address the learning and memory issues considered here.

2.2 THE OMNI DATA SET

The dataset we consider, part of the NSF-funded “Optimizing Memory using Neural Information” (OMNI) project[†], consists of human performance on a cued-recall memory test for a set of Lithuanian-English word translations. The learner’s task is to study the word pairs across multiple presentations and then, after a delay, recall the English associate for a presented Lithuanian word.

*Here we focus on fMRI due to improved spatial resolution, even though other methods (e.g., EEG and skin conductance response), also provide useful signals that correlate with memory performance and could be incorporated into our approach.

[†]<http://gureckislab.org/omni>

Starting with a normed set of Lithuanian-English words, we selected 45 translation pairs (Grimaldi et al., 2010). During study, participants saw the translation pairs presented one at a time for 4 seconds each with a variable duration inter-trial interval (4s-16s for consistency with event-related MRI timing). Words were presented on a computer screen with the Lithuanian word at the top of the screen and the English translation underneath.

Each word pair was presented five times and no pair was presented for the n th repetition until all words had $n - 1$ presentations. Importantly, and in contrast to many psychology studies on the subsequent memory effect, all participants see the same sequence of study items[‡]. Immediately following the study session participants gave judgments of learning (JOLs, (Nelson and Dunlosky, 1991)): for each pair participants were presented with the Lithuanian and English word and used the computer mouse to indicate on a scale of 0-100 how likely they were to remember the association in one week.

Participants were given either an immediate recall test (0 hours) or returned to the lab approximately 24, 72, or 168 hours after the initial study session (randomly assigned)[§]. During the recall test, participants saw a Lithuanian word presented on the screen and had to type the associated English word. A trial was coded as correct if participants typed the correct English word (allowing for typographic errors) and all other responses were incorrect.

For more efficient estimation of the different model parameters, we conducted a large

[‡]Although the models we apply do not explicitly model inter-item interactions, maintaining a fixed sequence across participants ensures that some of these inter-item effects will be captured in the model parameters we estimate because, for instance, the measured difficulty of a word is always assessed with respect to the other list items.

[§]Due to schedule difficulties a one subject returned at 48 hours but we still included their data in the modeling. In addition, 9 of the 72 hour subjects were scanned in a different fMRI scanner but we only include their behavioral data here.

behavioral experiment outside of the MRI scanner and combined those data with additional observations from participants who performed the same task during MRI scanning (under this view all participants are equally useful but purely behavioral subjects are treated as though their MRI data are “missing” and so estimates of their learning are based on the observed JOLs and recall performance). Each participant ($N=189$) was tested at one of the four study-test delays. Among the behavioral participants (i.e., no MRI data) the group N s were 20, 49, 60, and 49 in the 0, 24, 72, and 168 hour study-test delay groups, respectively. All MRI participants ($N=21$) were tested at the 72 hour delay.

MRI participants underwent an identical study-test procedure as the behavioral participants except they were scanned during the study session. MRI data were collected on a Siemens Prisma 3 T at the New York University Center for Brain Imaging. Functional Blood Oxygen-Level Dependent (BOLD) data covering the cortex were acquired at a spatial resolution of 2.5 mm^3 with a 1 second repetition time (TR; the temporal resolution of the fMRI data) and anatomical scans were collected at a spatial resolution of $.75 \text{ mm}^3$.

To summarize, the final data set consists of a record for each learner that contains: the pattern of recall attempts for each list item, JOLs collected after the study session for each list item, and, for each MRI participant, the $65 \times 77 \times 73$ set of voxel measurements across 2936 time-points describing the BOLD signal recorded with MRI.

Figure 2.1 shows key features of the behavioral data. Across the four different test delays, memory performance generally drops, likely due to forgetting. Participant performance varied widely from 0 to 100 percent correct. In addition, across participants, average JOLs following study were weakly correlated with performance ($r = [0.43, 0.24, 0.31, 0.55]$ and $p = [0.06, 0.10, 0.004, 3.4e-5]$ in the 0h, 24h, 72h, and 168h groups, respectively).

Pooling across all participants, the mean JOL correlation with final performance is low but significant, $r = .365, p < 1e-7$.

2.3 INFERRING KNOWLEDGE STATES FROM BEHAVIORAL AND NEURAL DATA

The following section describes the basic mathematical structure of our models. Similar to BKT, the core of our approach assumes a probabilistic representation of the latent mnemonic status (e.g., *remembered* versus *forgotten*) of each item on the to-be-remembered list and we begin with established two- and three-state models that have shown effectiveness in tracking learning and memory (Atkinson, 1972; Corbett and Anderson, 1995). Where our models differs from past knowledge tracing approaches is that we propose a mapping between these latent mnemonic states and patterns of brain activity that can allow the brain data to inform this inference.

2.3.1 A HIDDEN MARKOV MODEL OF MEMORY

Like BKT, our approach draws heavily from the structure of HMMs. Each memory trace, i , (i.e., memory for the association between two words) is represented as a non-homogenous, censored Hidden Markov Model with the following properties (notation follows (Rabiner, 1989)):

STATES

Each trace can be in one of a number of discrete mnemonic **states**, S . For simplicity we will begin with a two state $S = \{s_U, s_K\}$ model with states corresponding to *unknown* and *known* similar to BKT. However, we also consider a more complex, three-state model first

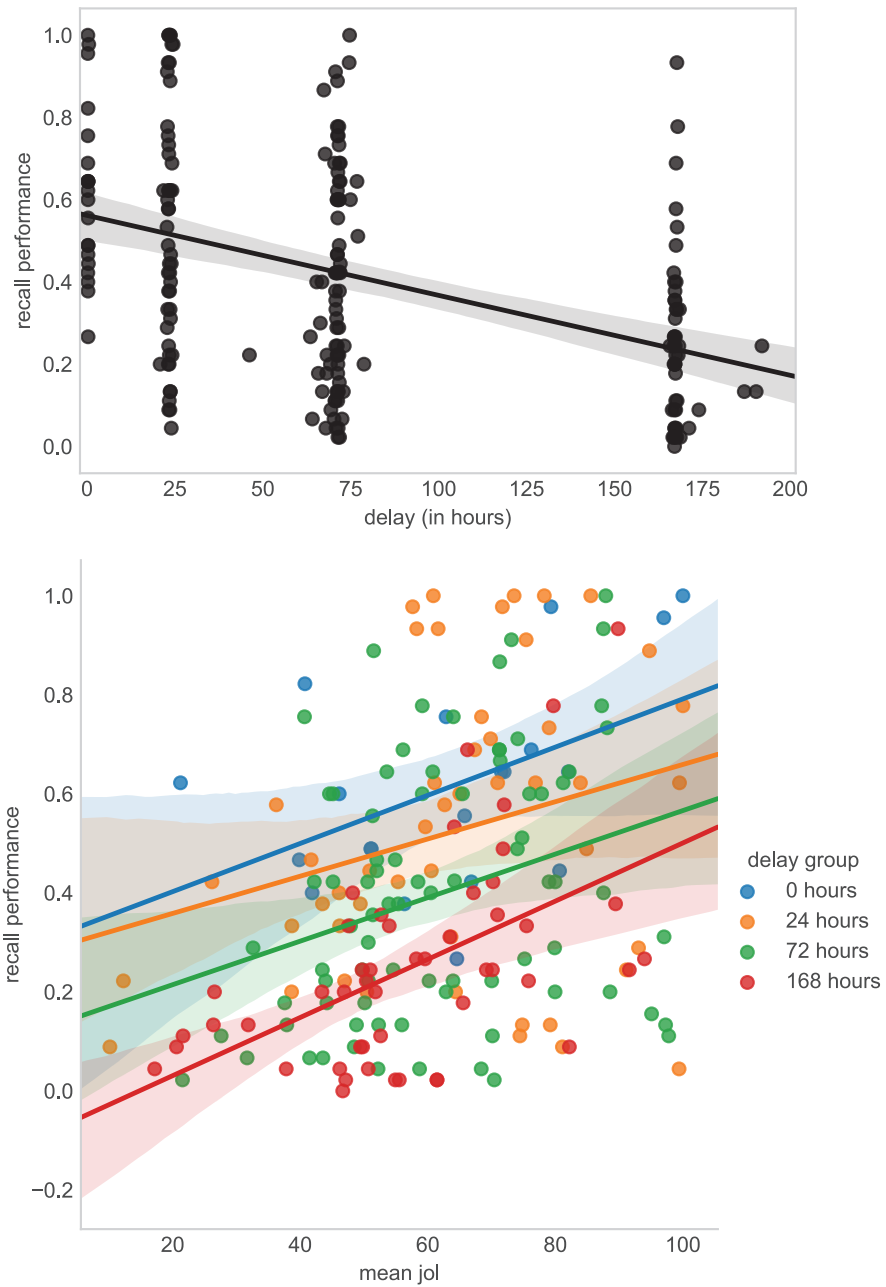


Figure 2.1: Top: Mean recall performance (% correct) for individuals (dots) at each study-test delay. Bottom: Mean individual participant Judgment of Learning is correlated with individual overall percent recalled within each delay condition.

proposed by Atkinson (Atkinson, 1972). The three-state model has states $S = \{s_U, s_K, s_P\}$ corresponding to *unknown*, *known* (with possibility of forgetting), and *permanently known* (see Figure 2.2). Across both types of models the s_K and s_P states represent memories that have generally higher recall probabilities (e.g., $\Pr[\text{recall} = \text{correct} | s_P] > 0$), but the s_K state is susceptible to decay between study events while the s_P state is absorbing[‡]. The current state of item i at time t will be denoted q_t^i .

PRIORS

A **prior**, $\pi_{t=0}$, that captures our initial belief of the memory state of all items. The prior for a particular item memory, i , can be written as $\pi_{t=0}^{i,s} = \Pr[q_{t=0}^i = s]$ for $s \in \{s_U, s_K\}$ (two state) or $s \in \{s_U, s_K, s_P\}$ (three state). With unfamiliar learning materials we assume that the initial memory status is heavily biased towards the unknown state (i.e., $\pi_{t=0}^{i,s_U}$ is much higher than for any other state).

TRANSITIONS

A set of **transition probabilities**, $\mathcal{A}$, which determine the likelihood that a memory will move between the different states at each time point. In prototypical HMMs the transition probabilities are stationary and the same transitions are applied at each time step. In our model there are different sets of transition probabilities which are applied at a given time step depend on the type of external “event”, $\mathcal{E}_t^i$, that occurs (e.g., a study trial versus a time step between trials; Figure 2.2). For memory trace i the transition probability of moving

[‡]One way for the model to capture the difference in performance at 24 versus 168 hours is to assume different mixtures of the s_K and s_P states following learning. For example, at 168 hours, traces in s_P state may dominate correct responses.

from state s to s' after an event of type g will be denoted $a_t^{i,g,s \rightarrow s'} = \Pr[q_t^i = s' | e_t^i = g, q_{t-1}^i = s]$ where g indicates the specific event type on trial t .

Event types depend on the particular experiment design but here include “study trial” (*study*), “study with JOL trial” (*study+JOL*), “timestep in which memory decays” (*decay*), and “test trial” (*test*). Generally, during *study* or *study+JOL* events we assume that items tend to transition from a more poorly learned state to a more fully learned state. The probability of transitioning to a new state on a study trial is represented in our three state model by parameters x, y and z and in the two-state model by parameter l (see Figure 2.2). During *decay*, items in a non-permanent state (s_K) have a probability of transitioning to the *unknown* state with probability f while items in s_P (in the three-state model) remain in the permanently learned state. Decay events are necessary to account for the patterns of forgetting across the study-test delay intervals shown in Figure 2.1. We assume test trials have no effect on transitions as they appear at the end of the task.

We define an experiment **protocol**, E , as a $\mathcal{N} \times \mathcal{T}$ matrix where $\mathcal{N}$ is the number of items being studied and $\mathcal{T}$ is the total number of micro-time steps modeled in the experiment. Each entry of the matrix, e_t^i , codes which of a discrete set of event types occurred on a time step as described above. The protocol captures the dependencies between event sequences that influence different memory traces. For example, if word w is studied on a given trial, then all the other items on the list might undergo a memory decay event during the same time step. This way the protocol enforces the implicit tradeoffs of studying one item over others at a particular point in time.

Two-state model

Study Transitions



Decay Transitions

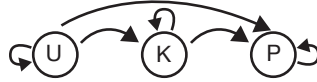


		q_{t+1}	
		S_U	S_K
q_t	S_U	$1-l$	l
	S_K	0	1

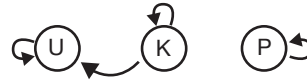
		q_{t+1}	
		S_U	S_K
q_t	S_U	1	0
	S_K	f	$1-f$

Three-state model

Study Transitions



Decay Transitions



		q_{t+1}		
		S_U	S_K	S_P
q_t	S_U	$1-y-z$	z	y
	S_K	0	$1-x$	x
	S_P	0	0	1

		q_{t+1}		
		S_U	S_K	S_P
q_t	S_U	1	0	0
	S_K	f	$1-f$	0
	S_P	0	0	1

Figure 2.2: The matrix of transition probabilities for either study or decay events in the two and three state model. The letters within each matrix reflect the transition parameters which are estimated to data. The state labels U are “unknown”, K are “known” (with possible forgetting), and P are “permanently known.”

OBSERVABLE SIGNALS

The mapping between brain and behavior is made through a set of **observation distributions**, B , which define the probabilities that, on event type g at time t , an observable random variable of data type d , $\mathbf{o}_t^{g,d}$ takes on a value $v_k^{g,d}$ from a (potentially infinite) alphabet $v^{g,d}$. For each memory trace i , we can write the probability of its associated observables as $b_t^{i,g,s,d}(v_k^{g,d}) = \Pr[o_t^{g,d} = v_k^{g,d} | e_t^i = g, q_t^i = s]$. Observation distributions in effect define the full generative model that links both behavior and neural information to underlying knowledge states. Here we consider three types of observations: behavioral assessments (*recall*), JOLs (*JOL*), and hemodynamic fMRI measurements (*fMRI*). However, this approach can easily incorporate many other measures including response time, pupil dilation, EEG measurements, or alternative fMRI signals.

Behavioral Assessments. At certain points during the experiment the protocol might define a memory test event. On these types of trials the subject might be asked to recall a studied item from memory or to recognize it from a list of alternatives. The response given on these trials is treated as an observation associated with this particular type of event. Specifically, the alphabet is $v^{test,recall} \in \{correct, incorrect\}$ and $v^{g,recall} \in \emptyset$ for $g \neq test$, reflecting the absence of any recall response on non-test events. The distribution of test question answers about memory trace i from state s at time t , is then $b_t^{i,test,s,recall}(correct) = p_{recall_s}$ and $b_t^{i,test,s,recall}(incorrect) = 1 - p_{recall_s}$ where p_{recall_s} is defined (or fitted) for each memory state. For other trial types, i.e. $g \neq test$, $b_t^{i,g,s,recall}(\emptyset) = 1$. So the update to state posterior probabilities on those events is driven by the state transitions. The parameters governing the probability of issuing a correct response conditioned on the latent memory state are equivalent to the “guess” and “slip” parameters in BKT.

Judgments of Learning. JOL responses were only given on the last study trial (a *study+JOL* event). JOL data were included in the model as the raw response/100 to each JOL trial for each person, i.e. $v^{study+JOL,JOL} \in [0, 1]$ and null for other trial types. We model the distribution of JOLs as a truncated Gaussian distribution in the range 0 to 1, i.e. $b_t^{i,study+JOL,s,JOL} = TN(\mu_{JOL_s}, \sigma_{JOL_s}, 0, 1)$ with μ_{JOL_s} and σ_{JOL_s} defined independently for each state s .

Hemodynamic fMRI measurement. Functional MRI scans provide time-series data for each of a large set of 3-dimensional voxels tiling the imaged volume (e.g., the brain). In studies measuring fMRI activation levels at specific timepoints it is common to estimate the activation level within voxels and then average voxels within spatial clusters, whether spatially contiguous (regions of interest, or ROIs) or sets of spatially disjoint but functionally related voxels showing similar response profiles (e.g., independent components). Due to the central limit theorem we can expect that the mean activation within a set of such voxels will be approximately normal. We also expect, based on prior work, that there will be a mean shift in the fMRI activation levels of various brain regions during study trials that are later remembered compared to those that are later forgotten (Davachi et al., 2003; Ranganath et al., 2003). We collect fMRI data for each *study* trial. The fMRI observation consists of N_{fMRI} features. Therefore, $v^{study,MRI} \in \mathbb{R}^{N_{fMRI}}$ and null otherwise. We model the fMRI state observation distributions as independent Gaussians for each feature n_i , i.e. $b_t^{i,study,s,MRI_{n_i}} = N(\mu_{MRI_{n_i,s}}, \sigma_{MRI_{n_i,s}})$.

INFERENCE

The full model is specified by a protocol, E , a set of priors over the states, $\pi_{t=0}$, a set of transition probabilities, A , and a set of observation distributions associated with each state-

event pair, B . Using Bayes' rule, the posterior probability that a memory trace on trial t is in state $s' \in S$ is:

$$\pi_t^{i,s'} = \frac{b_t^{i,g,s',d} a_t^{i,g,s \rightarrow s'} \pi_{t-1}^{i,s}}{\sum_j b_t^{i,g,s_j,d} a_t^{i,g,s \rightarrow s_j} \pi_{t-1}^{i,s}} \quad (2.1)$$

ILLUSTRATIVE CALCULATION

To illustrate the impact of hypothetical fMRI observations, consider Figure 2.3 which shows the protocol, E , for the timing of study events for two memory traces (Panel A): item 1 (black) and item 2 (white). On time points where item 1 is studied the protocol has a black cell (and similarly for item 2 using white). Panel B shows hypothetical evolutions over time for the two-state posterior probabilities $\{s_U, s_K\}$ for item 1 obtained by applying the study and forgetting transitions as shown in Figure 2.2 but without other observable information (i.e., a Markov model). In this example we set the l transition parameter applied on study events to 0.4 and the f parameter governing decay to 0.1.

At time point 1 the priors reflect the fact that before any study attempts a person is unlikely to know the item (e.g., $\pi_{i=0}^{i=1,s_U} = .9$). At time point 6, item 1 is presented for study for the first time and the posterior probabilities of each state are updated by applying the study transition probabilities to the state posteriors on time $t - 1$. Immediately after this study event, Panel B shows that there is now an increased probability of item 1 being in state s_K (solid orange line). However, between time point 6 and 40, item 1 is not presented again and so for each time step between we apply the decay transitions leading to gradual forgetting.

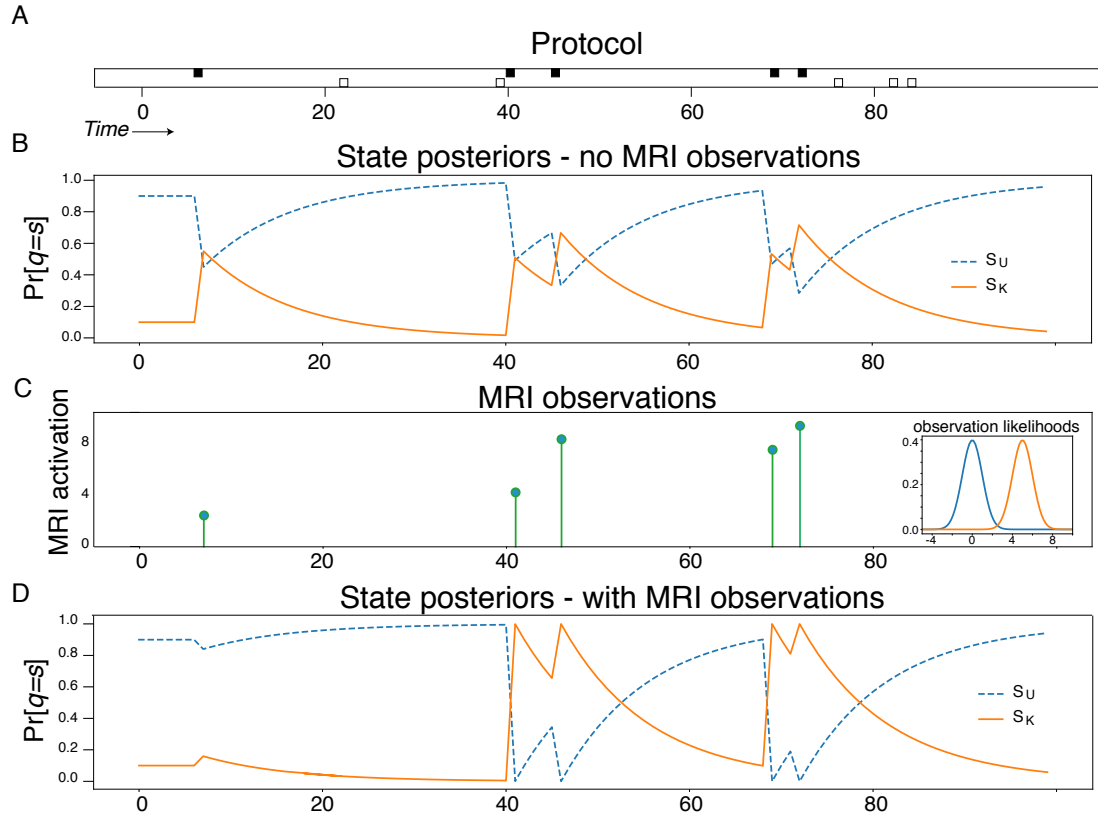


Figure 2.3: Example illustration of the effect of fMRI observations on inferences about latent knowledge in a two state-model. A) Protocol showing the timing of study events for item 1 (black boxes) and item 2 (white boxes). B) State posterior estimates for item 1 obtained from a hypothetical setting of the two-state model parameters (dashed blue = S_U , solid orange = S_K). C) Hypothetical “observed” fMRI signal on each study trial for item 1 (inset shows the probability density function over MRI observation values conditioned on the state). D) State posteriors for item 1 after incorporating the observation likelihoods from study trials for this item. The inferred state probabilities are dramatically altered by the incorporation of the MRI observation (see text).

The addition of observable signals that are probabilistically related to latent memory states alters these predictions. The inset figure in Panel C shows how the mean response from a set of voxels in the human brain might result in Gaussian-distributed summed BOLD signals that overlap but differ depending on the state of the memory (e.g., signal being stronger for s_K , orange, than for the s_U , blue, state). Panel C illustrates a hypothetical sequence of fMRI measurements that could be made about item 1 during the study trials (i.e., samples from the Gaussian distributions from the inset plot).

Panel D shows the posterior estimates of item 1's state at each time point obtained through combination of the transition dynamics **and** MRI observations (i.e., using the Hidden Markov Model). As can be seen comparing panel B and D, the addition of observations that are probabilistically associated with latent states can lead to different inferences about the posterior probabilities over those states. Until item 1 is presented at time point 6 the posterior estimates are the same in the Markov and Hidden Markov Models. However, at time point 6 we observe a fMRI signal of a particular magnitude which in turn has a likelihood of originating from each of the two underlying states. If we take into account the observed signal, our estimates of the posterior over states change, since a fairly small signal was observed and the likelihood of such a signal is substantially larger for state s_U than s_K . Consequently, our belief that the item is in state s_K is lower when we include the observation in our estimates than when we simply use the transition probabilities.

Similarly, at time point 40 item 1 is presented for a second study opportunity. Without observations our best estimate of the state probabilities suggests we should be indifferent between s_U or s_K , but the larger MRI observation observed is unlikely to have emerged from the unknown state and so the observation-constrained posterior estimates are weighted

much more heavily towards the s_K state. By including the Markov dynamics characterizing the likely temporal evolution of memories, we can adjudicate between otherwise ambiguous neural signals by appropriately dealing with uncertainty in measurement.

MODEL EVALUATION AND FITTING PROCEDURE

The following section details the model evaluation, comparison, and feature selection strategies we used.

Model parameterization. Partially due to identifiability concerns (van De Sande, 2013; Feng, 2017), some parameters were fixed to semantically coherent values (Doroudi and Brunskill, 2017), while others were estimated from the data.

For all words we fixed the initial state priors, $\pi_{t=0}$, as $[\cdot.99, \cdot.01]$ or $[0.99, 0.005, 0.005]$ for s_U, s_K in the two-state model or s_U, s_K , and s_P in the three-state model, respectively. This was motivated by the fact that none of the learners in our dataset had prior experience with Lithuanian. We also fixed the probabilities of giving the correct test response, $\mathbf{p}_{\text{recall}}$ as $[\cdot.01, \cdot.9]$ and $[\cdot.01, \cdot.9, \cdot.9]$ for latent memory states s_U and s_K (two-state model) or s_U, s_K , and s_P (three state model, see below), respectively. This reflects the assumption that it is very unlikely that one would guess the correct answer in a cued recall test without any memory ($s = s_U$) and that, as in (Atkinson, 1972), the primary difference between s_K and s_P in the three-state model is the susceptibility to decay over time rather than the availability of a memory to recall (via the influence of the f parameter; see Figure 2.2).

Fitted parameters include those determining the transition probabilities and observation distributions within each model. Both the two- and three-state models have transition probabilities to fit for each word pair w (summarized in Figure 2.2). In the two-state

model these are the l_w and f_w parameters controlling memory strengthening and decay, respectively. For the three-state models, the x_w , y_w , and z_w values control transitions between states during study opportunities and the f_w parameter determines forgetting rates.

Although the learning trajectories for each word pair were instantiated in separate HMMs, to get better estimates of the parameters we used a hierarchical Bayesian model that used group-level priors over the parameters to regularize the estimates. Each x_w was drawn from a Logit-Normal(x , σ_x) where x itself was drawn from a Normal(0, 6) and σ_x was drawn from a Truncated-Normal(0, 1). The model for the f_w parameters was exactly the same. The simplices zy_w were generated using the following procedure: z and y were drawn from a Normal(0, 6). z_w and y_w were drawn from Normal(z , σ_z) and Normal(y , σ_y) respectively with σ_z and σ_y both drawn from a Truncated-Normal(0, 1). Finally, zy_w was set to $\text{softmax}([0, z_w, y_w])$. This can be thought of as a multivariate generalization of the Logit-Normal with a diagonal covariance matrix.

When fitting models that incorporated JOLs or MRI data we also estimated the means and variance parameters for the Gaussian (truncated for JOLs) observation likelihood from each latent state. For the JOL distributions, each μ_{JOL_s} was drawn from a Normal(.5, .5) and each σ_{JOL_s} was drawn from Inverse-Gamma(1, 2). Similarly, for each fMRI feature n_i (see below) in state s , $\mu_{MRI_{n_i}s}$ was drawn from a Normal(0, 1) and $\sigma_{MRI_{n_i}s}$ was drawn from an Inverse-Gamma(1, 2).

fMRI feature selection. After standard MRI preprocessing (Danker et al., 2016), we selected data for inclusion in the model. We reduced the dimensionality of the fMRI data using group spatial independent components analysis (ICA) using the ICASSO algorithm as implemented in the GIFT ICA toolbox (<http://mialab.mrn.org/software/gift/>) (Calhoun

et al., 2001; van Maanen et al., 2011). This procedure, which is blind to trial information and memory outcome, resulted in a set of 60 independent components that are characterized by a particular temporal (the timecourse of activation) and spatial (the loading of each component on fMRI voxels) profile for each participant. Components that were unstable across estimations (ICASSO) and components associated with signal from ventricles or motion were discarded leaving 43 independent components for inclusion as model features. Individual trial activations for each identified component were summarized as the mean of timepoints encompassing 4-6 seconds post-stimulus onset (to account for the temporal lag in the BOLD response), resulting in one activation value for each trial in each component for each MRI participant.

Model estimation. We used MCMC sampling via the NUTS algorithm as implemented in Stan (Carpenter et al., 2017) to estimate the posterior over the parameters (4 chains of 200 iterations; 100 per chain discarded as burnin; 400 total samples per parameter). To ensure convergence, we checked that estimates of the probability of recall had low $\hat{R}$ values (a measure of whether the sampling chains are converging to similar estimates) (Carpenter et al., 2017; Gelman and Rubin, 1992).

Model evaluation. In order to compare models, we want to evaluate how well our models will predict new, unseen data. It is generally agreed that the generalization method with the fewest assumptions is leave-one-out cross validation, which is preferred when sufficient data and computational resources are available (Vehtari et al., 2017). To conserve on computational resources, here we use K-fold cross validation, setting K to 10. Because our goal is to assess the utility of incorporating MRI signals into a memory model, the held-out data only included data from the 20 fMRI subjects. We divided up the data from these subjects

into ten equally sized folds. We then trained ten versions of each model where the training set consisted of all of the data from behavior-only subjects and nine of the ten folds of the fMRI subjects. On the held-out test set, we used the identity of the words and the trial timings (and JOL or fMRI observations, where appropriate) to generate the posterior probability of recall for each held out word at the time of test.

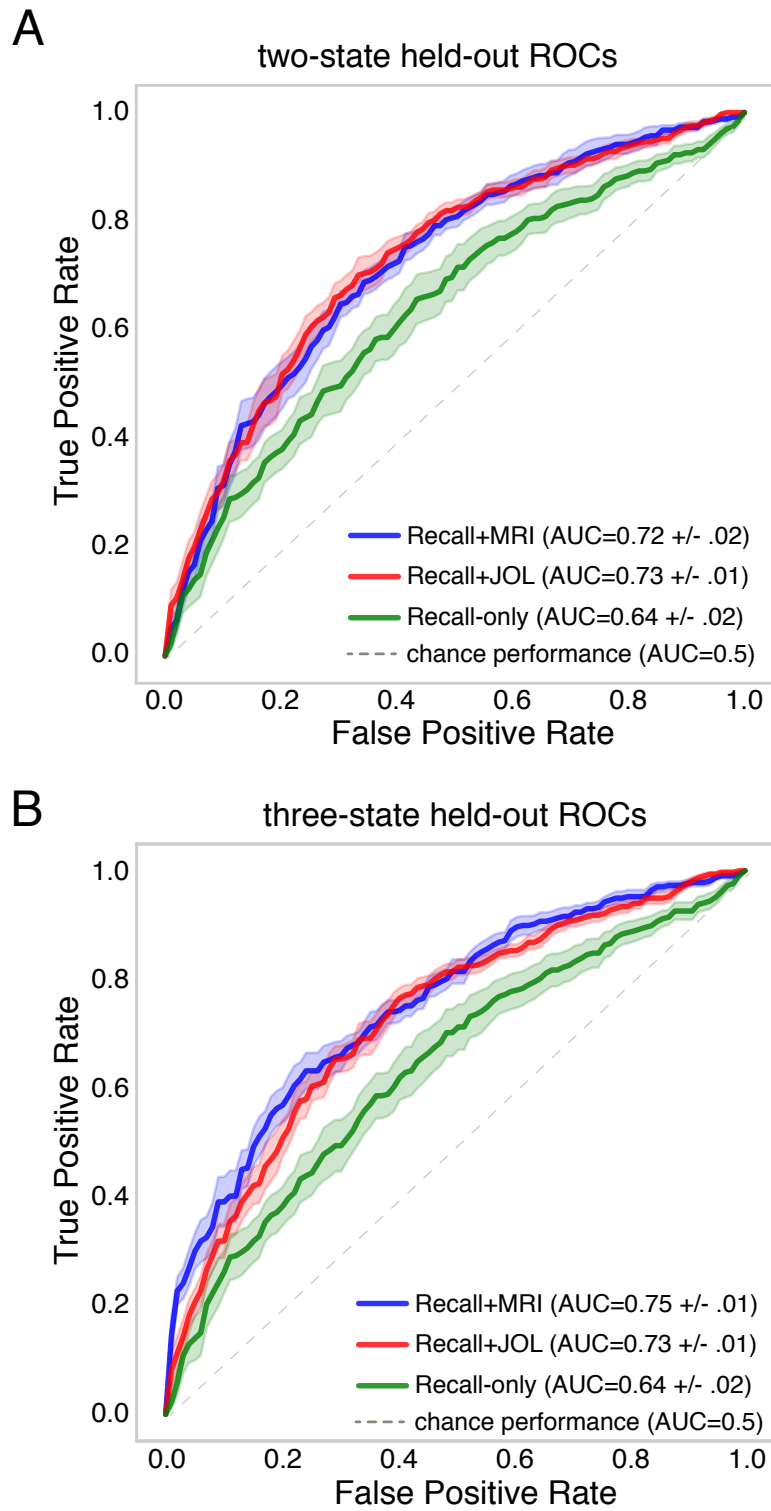
As we are primarily interested in our ability to classify a new piece of data as successfully recalled or not rather than the log likelihood of the trial under the model, we adopted a cross-validated area under the ROC curve metric (ROC-AUC). The ROC-AUC can be interpreted somewhat like an accuracy measure where 0.5 represents chance prediction and higher values indicate better predictive performance of the model. Using ROC-AUC allows us to compare the held-out predictive performance of models with varying numbers of parameters while providing a metric of model performance that is relatively insensitive to class imbalance and does not prioritize one kind of error over another (e.g., trading off Hits versus Misses). The model ROCs were defined by calculating, in each cross validation fold, the proportion of predicted as remembered trials that were recalled correctly (*Hits*) and the proportion of predicted as remembered trials that were not (*False Alarms*) at each level of posterior recall probability given by the model.

Model Comparison. We fit three variants of each of the two- and three-state models: a *Recall* model fit to trial timing and recall performance (the binary recall success scores for each word); a model fit to trial timing, recall performance, and JOL observations (*Recall+JOL*); and a model fit to trial timing, recall performance, and fMRI observations (*Recall+fMRI*). In each case the training data included data from all of the behavioral participants and a subset of the MRI participant data, and models were evaluated on held-out

Table 2.1: Cross validated Area Under the Curve of the Receiver-Operating Characteristic (ROC-AUC) with $\pm$ standard error (in parentheses) across folds.

	two-state model	three-state model
Recall	0.64 (.02)	.64 (.02)
Recall+JOL	0.73 (.01)	.73 (.01)
Recall+MRI	0.72 (.02)	.75 (.01)

data. The logic of these comparisons is to see if the models incorporating additional observations (*Recall+JOL* and *Recall+MRI*) provide a better basis for prediction than do the purely behavioral models. In addition, we are interested in whether the model incorporating MRI observations is able to outperform the model incorporating JOLs. This would suggest that the brain data contains more information relevant about memory performance than do people’s own self-reports about their memory fidelity. While we are ultimately interested in held-out predictive performance, the models do differ in model complexity. In raw numbers, for the two-state models, the *Recall* model had 2×45 word parameters and 4 hyperparameters, the *Recall + JOL* model added 4 parameters, and the *Recall + MRI* model added $4N_{\text{fMRI}}$ parameters. For the three state models, the *Recall* model had 4×45 word parameters and 7 hyperparameters, the *Recall + JOL* model added 6 parameters, and the *Recall + MRI* model added $6N_{\text{fMRI}}$ parameters. However, due to the hierarchical nature of these models, the effective number of parameters may have differed depending on the amount of regularization done by the hierarchical prior.



56
Figure 2.4: ROC curves for held-out predictions in each of the two-state (panel A) and three-state (panel B) model variants (*Recall*, *Recall+JOL*, *Recall+MRI*). The curves show the mean $\pm$ sem across each of the cross validation folds.

2.4 RESULTS

2.4.1 TWO-STATE MODEL

For each variant of the two-state model (*Recall*, *Recall+JOL*, *Recall+MRI*) we computed the ROC-AUC for predictions of recall accuracy in held-out trials for the MRI participants. The *Recall* model, trained on the timing of study and test trials and recall performance, achieved a mean (across held-out folds) ROC-AUC of 0.64 (± 0.02), providing an above chance baseline model against which to evaluate the utility of JOL and fMRI observations (Figure 2.4A).

The *Recall+JOL*, which adds judgments of learning to both the training and evaluation of the *Recall* model, achieved a mean held-out ROC-AUC of 0.73 (± 0.01), improving our predictions relative to the *Recall* model. This shows that metacognitive judgments collected from individuals at the end of a learning session can be used to refine predictions about held-out recall performance.

We next assessed whether fMRI signals recorded during study events could be leveraged to make predictions about held-out performance. The *Recall+MRI* model yielded a held-out ROC-AUC of 0.72 (± 0.02). Although the held-out performance did not surpass the *Recall+JOL* model, this result indicated that there may be information in the MRI measurements that could be used to make predictions about held-out memory recall performance.

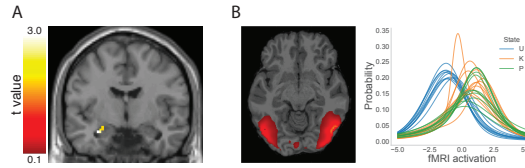


Figure 2.5: Examples of using estimated model to analyze the brain. A) Coronal slice showing left anterior hippocampal voxels tracking the change in s_K state posterior for each study trial. B) Topography (left; axial slice) and posterior predictive distributions (right) for MRI activations from most informative component in the three-state model. Individual traces show the distributions for each fold of the cross validation

2.4.2 THREE-STATE MODEL

We next considered whether a more elaborated model of memory could leverage more subtle dynamics of the fMRI data.^{||} The held out ROC-AUCs for the *Recall* and *Recall+JOL* three-state models did not differ from those observed in the two-state model (Figure 2.4B). However, the three-state MRI model boosted the held-out AUC to .75 ($\pm .01$) which was an improvement compared to the original two-state *Recall+MRI* model. This was also, in terms of held-out predictions, the most successful model we considered in these comparisons (but see Conclusions), building confidence in the utility of incorporating neural signals into knowledge tracing models.

In addition, whereas the *Recall* and *Recall+JOL* models did not discriminate between the two- and three-state models, the fMRI data enabled better predictions using the three-state model, highlighting the utility of neuroimaging data in selecting between cognitive models.

^{||}Although our primary interest in this work is evaluating the held-out predictions of our models, we note that complexity of the three-state model means that three-state *Recall* or *Recall+JOL* variants may not be identifiable due to the sparseness of observations (a single recall outcome or the recall outcome and a single JOL) (van De Sande, 2013; Feng, 2017) However, for the MRI participants we have data for every trial, enabling estimation of a three-state *Recall+MRI* model.

2.4.3 RELATING MODEL DYNAMICS TO THE BRAIN

In addition to the improvements in memory prediction afforded by joint modeling of behavioral and neural data, our approach also allows for examination of fMRI data in light of the estimated models. Figure 2.5 presents two example analyses in this vein.

Figure 2.5A shows the contrast map resulting from regressing the change in posterior probability of s_K associated with each study trial (as estimated in the two-state *Recall* model) against the fMRI time-series in each voxel. Using the estimated two-state *Recall* model parameters, we extracted the state posteriors on each study event for the MRI participants based on the sequence and timing of study trials. We then calculated the change in predicted state posterior from just before to just after a study trial and used this change as the predictor for brain activations. This analysis is related to the General Linear Model approach often used in the subsequent memory literature, except that rather than using binary regressors that coded for *remembered* or *forgotten* outcomes as determined by a recall test, we used the estimated continuous state posteriors from the two-state model.

Using a knowledge tracing model in this way to provide estimates of when a particular item is learned during a study sequence with multiple repetitions allows for more sensitive analyses of the brain’s relationship to cognitive processes unfolding over extended time. Interestingly, we found that the voxels significantly correlated with the change-in-state-posterior regressor were a cluster in left anterior hippocampus, consistent with the hypothesized role for this region in encoding new information into memory (Davachi, 2006).

An alternative way to use the fitted models is to examine the estimated fMRI features’ observation likelihoods for each latent knowledge state. The *Recall+MRI* model included activation from a number of independent components as candidate neural features. Af-

ter estimating the model, the fMRI observation parameters can be used to assess which components provided information about the latent model states. Used in this way, the joint model can be used as a tool for understanding how complex cognitive dynamics, especially those that might not be apparent in a more conventional analysis (e.g., a traditional subsequent memory analysis that only considers activation at the time of study and performance at the time of test), are instantiated in the brain. The most informative component in our model was associated with voxels in lateral occipital and fusiform gyrus regions involved in processing complex visual inputs, as shown in an axial slice through the brain (anterior/posterior of the brain is up/down in the image) in figure 2.5B. The posterior predictive distributions for component activation conditioned on model state are also shown in figure 2.5B, and these estimated distributions showed stronger activation for items in the K or P states relative to U.

2.5 CONCLUSIONS

We evaluate a framework for integrating neuroimaging recordings into a knowledge tracing model. Our approach builds upon recent reports showing robust memory-related signals in the brain. We collected a medium-sized data set of human participants performing a second-language acquisition task both inside and outside a scanner. We then compared a variety of models on their ability to predict held out data for the MRI participants. Our most predictive model was a three-state hidden Markov model that incorporated neural measurements. This is interesting because this model was more predictive than alternative approaches that leveraged participants' self-assessment of their learning (JOLs). One conclusion from this analysis is that there seem to be measurable signals in the brain that index

the quality of memory with higher fidelity than people’s own introspective access.

We also observed that the use of fMRI measurements enabled discriminating between models that were equivalent when using behavioral data (recall or JOL) alone. Whereas the held-out performance of the two- and three-state models was the same for the *Recall* and *Recall+JOL* model variants, using fMRI data to inform the model estimation revealed an improvement for the three- compared to the two-state model. This result points to the ways in which joint modeling of behavioral and neural data can afford insights into cognitive dynamics that might not be available to researchers focusing on more restricted kinds of data.

Although the results are promising, our assumptions about the fMRI data at this stage are simplistic. For example, our model assumed that the distribution of fMRI signals was stable across time. However, it is well known that fMRI signals often show a pattern of *repetition suppression* (Grill-Spector et al., 2006) where the measured BOLD signal is systematically lower on subsequent presentations of an item. A more sophisticated analysis of the brain may lead to improvements in our models. Another particularly interesting direction is to attempt to model individual learner abilities (c.f. Yudelson et al., 2013; Liu and Koedinger, 2017) on the basis of patterns of brain activity given the large variance in overall performance across participants (see Figure 2.1).

Modifications to the model structure might also improve predictions. As an example, in ongoing work we estimated the three-state *Recall+MRI* model but modeled the fMRI observations as arising from transitions between states rather than from the states themselves (i.e., each fMRI component has a distribution of activations associated with *staying* in a state and another distribution associated with *switching* between states). The three-state

version of this *Recall+MRI-Transition* model yielded a held out AUC of 0.77 ($\pm .02$), which is our best performing model to date. This shows that there is certainly more signal we can exploit from the data by improving our generative model of the fMRI signal. Attempts to improve the fMRI modeling and explore different model structures are continuing.

We have also illustrated several ways in which this kind of simultaneous modeling approach might feedback to our understanding of the role of the brain in supporting learning and memory. Using a model-based regressor coding for the change in posterior probability of latent knowledge states, we identified a significant effect in a left anterior hippocampus region that is known to be involved in memory formation on the basis of past studies (Davachi, 2006). The similarities between this novel analysis approach and past cognitive neuroscience studies give converging evidence about the hypothesized role of these regions. We also used our estimates of the fMRI observation distributions to examine the relationship between fMRI activation arising from different neural components and the latent knowledge states instantiated in the model(s), which is a novel approach to understanding the way psychological mechanisms or processes may be implemented in the brain.

While we acknowledge the practical limitations of acquiring neuroimaging data in an educational setting – although advances in EEG technology and the established ability to measure subsequent memory signals with EEG may enable such use in restricted settings (Fukuda and Woodman, 2015; Dikker et al., 2017) – overall we believe this work represents an encouraging first step for knowledge tracing approaches that utilize indirect neural information as opposed to explicit tests.

3

Getting Blood from a Stone: Improving Neural Inferences without Additional Neural Data

3.1 NEURAL INFERENCE

One factor which limits progress in cognitive neuroscience is statistical precision and power (Munafò et al., 2019; Cremers et al., 2017; Yarkoni, 2009). Many effects studied in cognitive neuroscience are small relative to the noise in neuroimaging signals, a problem compounded by the small sample sizes used in most studies. Within a hypothesis testing framework, a lack of precise estimates can make it difficult to replicate reported effects. Even quite strong, known effects like motor movements predicting neural activity in motor regions can be unlikely to reach significance in a small sample (Poldrack et al., 2017). Having lower precision also means that any estimated effect that is significant will also have a high probability of the estimate being much greater in magnitude or even having the wrong sign compared to the true effect (Gelman and Tuerlinckx, 2000; Cremers et al., 2017; Yarkoni, 2009). However, due to a combination of factors including lack of funding for any particular project or lab, lack of appreciation for the statistical consequences of low precision and low incentives for improving the situation, most studies in cognitive neuroscience continue

to be underpowered (Poldrack et al., 2017).

Many remedies have been proposed for this situation, with the most common being to simply collect more neuroimaging data (Poldrack et al., 2017; Munafò et al., 2019). Because the funding available to many labs is relatively small (compared to the cost of running a single subject in a neuroimaging study), this might require a move towards working in larger consortia. For an individual lab that is not involved in a consortium, the best option might be to make use of publicly available repositories of open datasets, which can allow for new re-analyses and meta-analyses (Poldrack et al., 2017). However, this is not the only option for dealing with a lack of precision. Recent research has taken many approaches to increasing the signal-to-noise ratio of neuroimaging measures including creating more detailed statistical models of the neuroimaging signal (Lindquist et al., 2009), improving experimental design (Durnez et al., 2017) or even improving the measurement process itself (Feinberg and Yacoub, 2012). In addition, it has been suggested that a focus on computational cognitive modeling and behavior can also allow us to extract more signal (Krakauer et al., 2017; Palmeri et al., 2017) by creating better models of the underlying cognitive and neural processes. In this paper we explore yet another potential option for improving the statistical precision of cognitive neuroscience studies.

We begin by noting that many common statistical analyses in cognitive neuroscience involve behavioral data, typically collected from the subjects who also provided neural data. In one common scenario, the interest is in the relationship between individual differences in behavioral traits and structural measures of neural activity, such as performance in a task and functional connectivity (Rosenberg et al., 2016). In another typical analysis, researchers are interested in how within-subject changes in a cognitive state variable over time

relate to changes in the neural signal from a particular brain region, such as the prediction error in a reinforcement learning task (O'Doherty et al., 2003). One interesting, but often overlooked, point is that these behavioral measurements (i.e. responses to questionnaires or other stimuli, estimates of task performance and estimates of cognitive model parameters) are themselves noisy estimates of underlying cognitive processes and distributions of cognitive processes in the population. Thus our ability to infer relationships is limited by sampling error (because a small sample cannot provide much information about the joint distribution of traits and neural measures in the population) and measurement error (because we use models fit to data in order to measure the values of latent variables). Ignoring these two sources of noise can prevent us from learning as much as possible from our expensive neural data.

In the following, we demonstrate that one way to reduce both sources of error and improve statistical estimates of the relationship between the brain and behavior is to collect behavioral data from subjects *without collecting corresponding neuroimaging data*. In the modern era of online data collection, behavioral data is often significantly less resource-intensive to collect than neuroimaging data, in terms of money and the time and effort of both researchers and human subjects. We show how using modern statistical methods that can leverage information from the additional behavioral data can help mitigate the two types of noise. This ability has implications for the design of cognitive neuroscience experiments. In particular, due to their relatively lower cost, it can actually be possible that the most effective use of resources is to collect more behavioral subjects rather than more neuroimaging subjects.

Many statistical analysis designs in cognitive neuroscience involve only relating objec-

tive features of the stimulus or task to the neural signal. For instance, researchers might use regression to compare signals on trials with large rewards offered vs. small rewards (Knutson et al., 2000) or trials with stimuli that involve faces vs. scenes (Kanwisher et al., 1997). Because behavior does not enter the analysis directly, our discussion here will largely not address these types of analyses.

For the purposes of understanding how behavioral data can help, it will be useful to classify some common types of analyses used in cognitive neuroscience and lay out some notation for the rest of the paper. While this is by no means an exhaustive list, we claim that many analyses can be understood as falling into one of these three categories.

Event-Related Analysis Which brain regions are responsive to certain stimuli (or features of stimuli)? For instance, is there a region whose activity is related to the color of a stimulus? Or a region whose activity depends on the relative values of the options being presented? To do this, we can compare neural signals when subject i presented with a set of stimuli s that occur on trials (or times) $t \in \{1..T_i\}$. Each stimulus s is usually decomposed into a vector of features $\mathbf{f}_s$. These features can range from just the identity of the stimulus (?) to objective features of the stimulus (the reward being offered, the color of the stimulus) to subjective features of the stimulus. We can then fit a regression predicting neural signals n_{it} from the different values of the stimulus features

$$n_{it} = \alpha_i + \beta_i \mathbf{f}_{sit} + \varepsilon_{it} \quad (3.1)$$

where ε is uncorrelated noise (for the purposes of this discussion, we ignore the hemodynamic response). Typically, researchers are interested in which neural signals have a non-zero β parameter. Many modern multivariate methods invert this regres-

sion and try to predict the stimulus from the neural signal or estimate the correlation between distance in a stimulus space and distance in the neural signal space. However, for our purposes we still classify these studies as investigating the relationship between the brain and features of stimuli. As mentioned above, when $\mathbf{f}_s$ involves only objective features, collecting additional behavioral data will not be of much use. However, if $\mathbf{f}_s$ involves subjective features, such as a behavioral response to the stimulus by subjects in the experiment (e.g. a valence rating (Chikazoe et al., 2014)) or average behavioral responses from an outside dataset (e.g. typicality ratings (Jordan et al., 2016)), some of the methods described below may be relevant. We return to this in a later section.

Model-Based Event-Related Analysis Which neural signals are related to certain latent cognitive state variables? For instance, is there a brain region that is related to the prediction error in a reinforcement learning model of decision making? (O’Doherty et al., 2003). Typically, in order to generate variation in the latent variables, we present subjects with different stimuli. We can compare neural signals when subject i is using a cognitive process $C(\theta, \mathbf{c}, \mathbf{s})$ where θ_i are latent cognitive “trait” variables that do not vary over the relevant time, $\mathbf{c}_{it}$ are the latent cognitive “state” variables that do vary over time and $\mathbf{s}_{it}$ are the stimuli that were presented to the subject. While we can’t observe $\mathbf{c}$ directly, researchers collect behavioral responses $\mathbf{b}_{it}$ and assume that they are generated from the cognitive model such that $\mathbf{b}_{it} \sim C(\theta_i, \mathbf{c}_{it}, \mathbf{s}_{it})$. We can then fit a regression predicting brain activity from the cognitive states

$$n_{it} = \alpha_i + \beta_{\mathbf{c}_i} \mathbf{c}_{it} + \varepsilon_{it} \quad (3.2)$$

Like in standard **Event-Related Analysis**, researchers are interested in which regions of the brain have a non-zero β parameter.

(Model-Based) Individual Differences Analysis Is there a relationship between individual differences in behavior and individual differences in structural measures of the brain? Researchers often use a direct measure of a subject's behavior b_i such as answers to a questionnaire (Treadway et al., 2013) or performance on a behavioral task (such as a sustained attention task (Rosenberg et al., 2016)). More recently (?), researchers have also used cognitive "trait" variables from a cognitive model fit to behavioral data (i.e. θ_i above). Structural measures can include measures like functional connectivity or average resting state activity that do not depend on time. In this setting, researchers are typically interested in characterizing the joint distribution of neural measures n_i and behavioral (b_i) or latent cognitive (θ_i) trait variables. In particular, if the neural measures and behavioral variables can be approximated by a bivariate normal, researchers want to find neural measures where the correlation ρ is non-zero.

3.2 TWO SOURCES OF ERROR IN NEUROIMAGING ANALYSES

First, we describe each of the two sources of error in more detail as well as how they can decrease the precision of inferences about the relationship between neural signals and behavior.

3.2.1 SAMPLING ERROR

In an **Individual Differences Analysis**, we want to know what ρ is in the general population but estimate it by using the sample correlation r . With a small number of subjects, r can be quite far from the true ρ . An approximate confidence interval for ρ can be constructed as $\text{arctanh}(r) \pm \frac{1}{\sqrt{s-3}}$, where s is the number of subjects or, more generally, the number of data points used to estimate r . As this makes clear, the precision of the estimate (and thus finding a ρ that we can say is non-zero with confidence) is constrained by the size of our sample. In many neuroimaging studies this is often relatively small because neuroimaging data is expensive and time consuming to collect. In Figure 3.1, we show an example of where the smaller sample size can lead to noisy estimates and larger error bars.

3.2.2 ESTIMATION ERROR

In the **Model-Based Event-Related Analysis** described above, we want to find a set of neural signals that have a linear relationship with latent cognitive states $\mathbf{c}$. The true model we are interested in is

$$n_{it} = \alpha_i + \beta_{c_i} \mathbf{c}_{it} + \varepsilon_{it} \quad (3.3)$$

where in particular we are interested in which neural signals have a significant β_{c_i} parameter (typically, at the group level). In practice, because we don't have direct access to $\mathbf{c}_{it}$ we typically replace it with estimates $\hat{\mathbf{c}}_{it}$ by fitting the model $C(\theta, \mathbf{c}_{it}, \mathbf{s}_{it})$ to the behavioral data b_{it} . It is typically assumed that the model doesn't perfectly account for behavior and that there is some variability in responses beyond what can be predicted by the model. For instance, when fitting a reinforcement learning model, a decision noise parameter is often used to

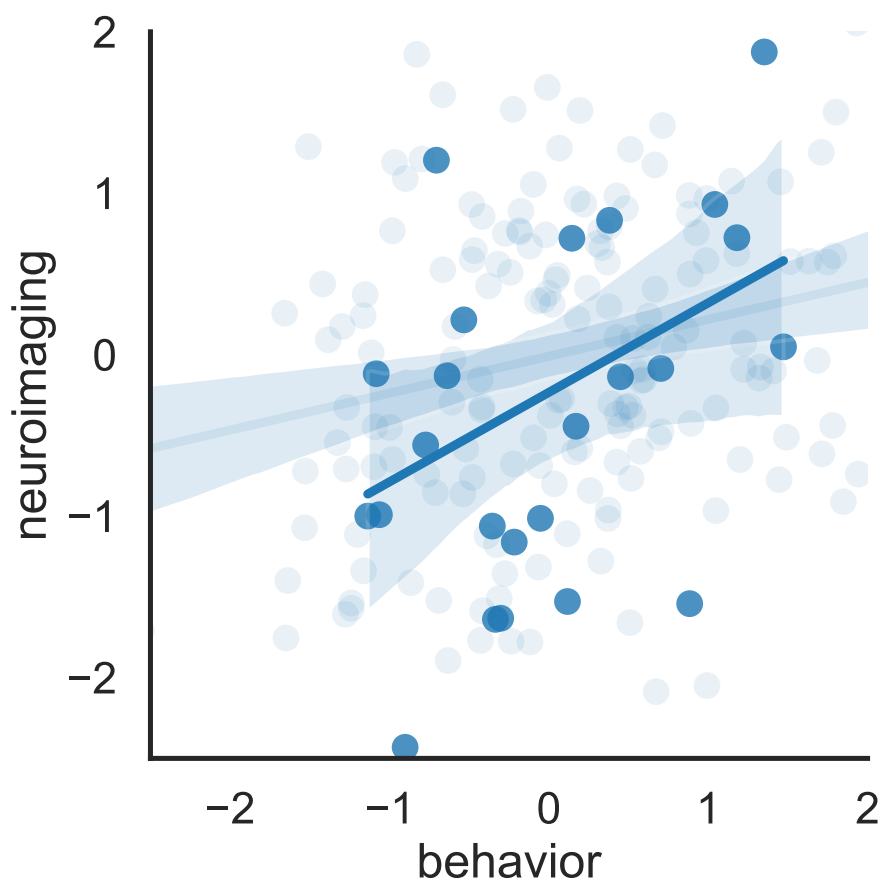


Figure 3.1: 25 samples from a multivariate Gaussian distribution are overlaid over an additional 100 samples from the same distribution. Regression lines are plotted showing how estimates from smaller samples can add significant variance to the estimated correlation. Error bars show that we cannot rule out a zero correlation in the smaller sample.

deal with variability beyond what can be accounted for by the latent representations in the model. Because we don't have infinite data, this typically means that several sets of parameters could account for the data even if the model is well identified. In other words, the parameters $\hat{c}_{it}$ are estimated with some uncertainty. This may seem obvious to many readers but standard techniques that are typically used in cognitive neuroscience do not easily allow this uncertainty to be taken into account in the above regression. Instead of the model above, researchers often simply plug in the estimated $\hat{c}_{it}$ into the regression, i.e.

$$n_{it} = \alpha_i + \beta_{\hat{c}_i} \hat{c}_{it} + \varepsilon_{it} \quad (3.4)$$

In this model, the value of the slope estimate using the estimated latent variables, $\beta_{\hat{c}_i}$, will not necessarily be the same as when using the true parameter β_{c_i} . In fact, it can be shown that on average, the least squares estimate of $\hat{\beta}_{\hat{c}_i}$ will be

$$E(\hat{\beta}_{\hat{c}_i}) = \beta_{c_i} \rho(\mathbf{c}_i, \hat{\mathbf{c}}_i) \quad (3.5)$$

where $\rho(\mathbf{c}_i, \hat{\mathbf{c}}_i)$ is the correlation between the true $\mathbf{c}_i$ and the estimates $\hat{\mathbf{c}}_i$ (Wilson and Niv, 2015). This is perhaps not always relevant since the neuroimaging signal can often only be known up to a multiplicative constant. However, Wilson and Niv (2015) also showed that, as long as the estimates are uncorrelated with the regression error term ε_{it} , the expected t -statistic for the $\hat{\beta}_{\hat{c}_i}$ is approximately

$$E(\hat{t}) = \beta_{c_i} \rho(\mathbf{c}_i, \hat{\mathbf{c}}_i) \sqrt{\frac{T_i - 2}{\sigma(\varepsilon)^2 + \beta_{c_i}^2 (1 - \rho(\mathbf{c}_i, \hat{\mathbf{c}}_i)^2)}} \quad (3.6)$$

where T_i is the number of trials from subject i used to estimate the regression. We can see that this t -statistic gets closer to 0 as $\rho(\mathbf{c}_i, \hat{\mathbf{c}}_i)$ gets further from 1. A smaller t -statistic means that the standard error (or any multiple of it such as a confidence interval) is more likely to overlap with 0. Thus having a lower correlation between the estimates and the true values, or, in other words, greater estimation error, can make it harder to find relevant signals in the typical regression setting.

Estimation error can also be relevant for **Individual Differences** analyses looking at correlations between latent cognitive trait variables and neural signals. If we estimate the trait variable θ_i from behavioral data, [Katahira \(2016\)](#) showed that the correlation between estimates $\hat{\theta}$ and structural neural signals $\mathbf{n}$, can be decomposed as

$$\rho(\mathbf{n}, \hat{\theta}) = \rho(\mathbf{n}, \theta) \rho(\theta, \hat{\theta}) \quad (3.7)$$

Therefore, the correlation is upper bounded by the true correlation between the neural signals and latent trait variables $\rho(\mathbf{n}, \theta)$ and will decrease as the correlation between the estimates and the true values $\rho(\theta, \hat{\theta})$ decreases (or as estimation error increases). Because the size of confidence intervals for a correlation coefficient only depend on the total number of subjects, the intervals for a correlation with a noisier estimate will be more likely to overlap with zero. This means that again, greater estimation error will make it harder to find relevant signals.

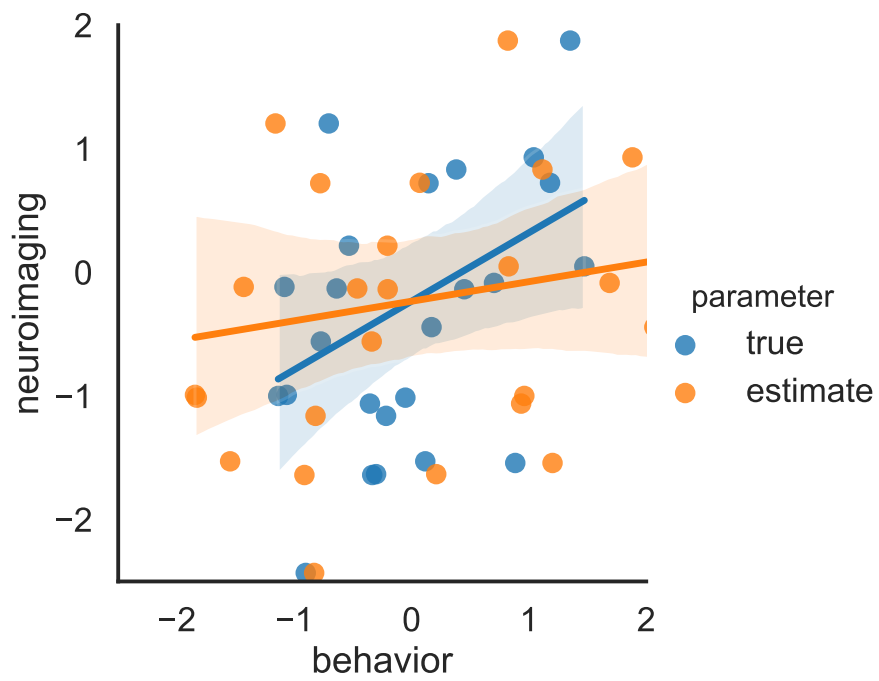


Figure 3.2: Here we show a case where the behavioral data is estimated with error. The correlation with the estimates is lower than it would be with the true behavioral data.

3.3 BEHAVIORAL DATA

Here we show some examples of how data collected from subjects outside the scanner can reduce the kinds of error described above and improve inferences about the relationship between neural signals and behavior.

3.3.1 BEHAVIORAL DATA AS NEUROIMAGING DATA WITH A MISSING VARIABLE

Typically we are interested in generalizing beyond our sample and characterizing the joint distribution of neural signals and behavioral variables in the population. However, as mentioned, most neurimaging studies are limited by getting a relatively small sample. One somewhat counterintuitive notion from the field of statistics focused on missing data is that we can often get better estimates of a parameter in the population (in terms of a lower mean squared error) by including data points that do not have every variable recorded in our analysis (Little and Rubin, 2019). This is because we are often assuming that variables have some joint distribution. If the correlation between two variables is not zero then we can know something about the value of what a missing variable must be from knowing the value of the other variable. If we have collected neural and behavioral data from s_{nb} subjects and only behavioral data from an additional s_b subjects. Anderson (1957) showed that the maximum likelihood estimates of the mean and standard deviation of $\mathbf{n}$ as well as the correlation between $\mathbf{n}$ and $\mathbf{b}$, ρ , are not the same when we include all of the data points as they would be if we only include the data with all variables recorded (the complete data). We can see this by rewriting the likelihood for a bivariate normal. If

$$(\mathbf{n}, \mathbf{b}) \sim N(\mu, \Sigma) \tag{3.8}$$

where $\mu = (\mu_n, \mu_b)$ and $\Sigma = \begin{pmatrix} \sigma_n^2 & \rho\sigma_n\sigma_b \\ \rho\sigma_n\sigma_b & \sigma_b^2 \end{pmatrix}$ then, we can factor the likelihood as

$$P(\mathbf{n}, \mathbf{b}) = P(\mathbf{n}|\mathbf{b})P(\mathbf{b}) = N(\mathbf{n}; \alpha_{nb} + \beta_{nb}\mathbf{b}, \sigma_{nb})N(\mathbf{b}; \mu_b, \sigma_b) \quad (3.9)$$

The advantage of writing the joint distribution as we did above is that the first term is just the regression of $\mathbf{n}$ on $\mathbf{b}$ which is only defined for the data points that have both variables recorded. Therefore, we can get the maximum likelihood estimate for α_{nb}, β_{nb} and σ_{nb} using standard least squares regression estimates with the n_{nb} data points. Maximizing the likelihood of the second term only depends on the $\mathbf{b}$ data so we can use standard estimates of the mean and variance using all of the data. We can write the correlation as

$$\rho = \frac{\text{Cov}(\mathbf{b}, \mathbf{n})}{\sigma_n\sigma_b} \quad (3.10)$$

Then plugging in the maximum likelihood estimates $\hat{\beta}_{nb}, \hat{\sigma}_{nb}$ and $\hat{\sigma}_b$, we get the maximum likelihood estimate for ρ using all of the data:

$$\hat{\rho} = \frac{\beta_{nb}\sigma_b^2}{\sqrt{\sigma_{nb}^2 + \beta_{nb}^2\sigma_b^2}} = \frac{\beta_{nb}\sigma_b}{\sqrt{\sigma_{nb}^2 + \beta_{nb}^2\sigma_b^2}} \quad (3.11)$$

Because the maximum likelihood estimate of σ_b using all $n_{nb} + n_b$ data points will not in general be the same as only using the n_{nb} data points, the maximum likelihood estimate of ρ using all the data will not be the same as using only the complete data.

We can now try to illuminate how this new estimate is different from the correlation estimate based on only the n_{nb} complete cases. [Little and Rubin \(2019\)](#) derive the missing

data estimator $\hat{\rho}$ in a different way, showing that

$$\hat{\rho} = r \left(\frac{\hat{\sigma}_b}{\tilde{\sigma}_b} \right)^{\frac{1}{2}} \left(\frac{\tilde{\sigma}_n}{\hat{\sigma}_n} \right)^{\frac{1}{2}} \quad (3.12)$$

where $r, \tilde{\sigma}_b, \tilde{\sigma}_n$ are the estimates of the correlation, the variance of the behavioral variable and the variance of the neural variable among the n_{nb} complete cases. $\hat{\sigma}_n$ is the variance of n after adjusting for the $\hat{\sigma}_b$ estimate based on all of the data. [Little and Rubin \(2019\)](#) derive the formula as

$$\hat{\sigma}_n = \tilde{\sigma}_n + \hat{\beta}^2 (\hat{\sigma}_b - \tilde{\sigma}_b) \quad (3.13)$$

We can instead write this in terms of r and rearrange terms, i.e.

$$\hat{\sigma}_n = \tilde{\sigma}_n + \frac{r^2}{\tilde{\sigma}_n} (\hat{\sigma}_b - \tilde{\sigma}_b) = \tilde{\sigma}_n \left(1 + r^2 \left(\frac{\hat{\sigma}_b}{\tilde{\sigma}_b} - 1 \right) \right) \quad (3.14)$$

Plugging this back in, we get

$$\hat{\rho} = r \left(\frac{\hat{\sigma}_b}{\tilde{\sigma}_b} \right)^{\frac{1}{2}} \left(\frac{\tilde{\sigma}_n}{\tilde{\sigma}_n (1 + r^2 (\frac{\hat{\sigma}_b}{\tilde{\sigma}_b} - 1))} \right)^{\frac{1}{2}} = r \left(\frac{\hat{\sigma}_b}{\tilde{\sigma}_b} \right)^{\frac{1}{2}} \left(\frac{1}{(1 + r^2 (\frac{\hat{\sigma}_b}{\tilde{\sigma}_b} - 1))} \right)^{\frac{1}{2}} \quad (3.15)$$

To our knowledge, we are the first to show that the [Anderson \(1957\)](#) estimate, $\hat{\rho}$, only depends on the correlation estimate in the complete cases, r and the ratio of the variance of the behavioral variable in the complete cases and the full data set, $\frac{\hat{\sigma}_b}{\tilde{\sigma}_b}$. In figure 3.3, we plot how $\hat{\rho}$ changes as a function of these two variables. In general, if the variance increases including the additional data, the correlation estimate increases and visa versa.

[Garren \(1998\)](#) showed analytically that asymptotically, the maximum likelihood estimate

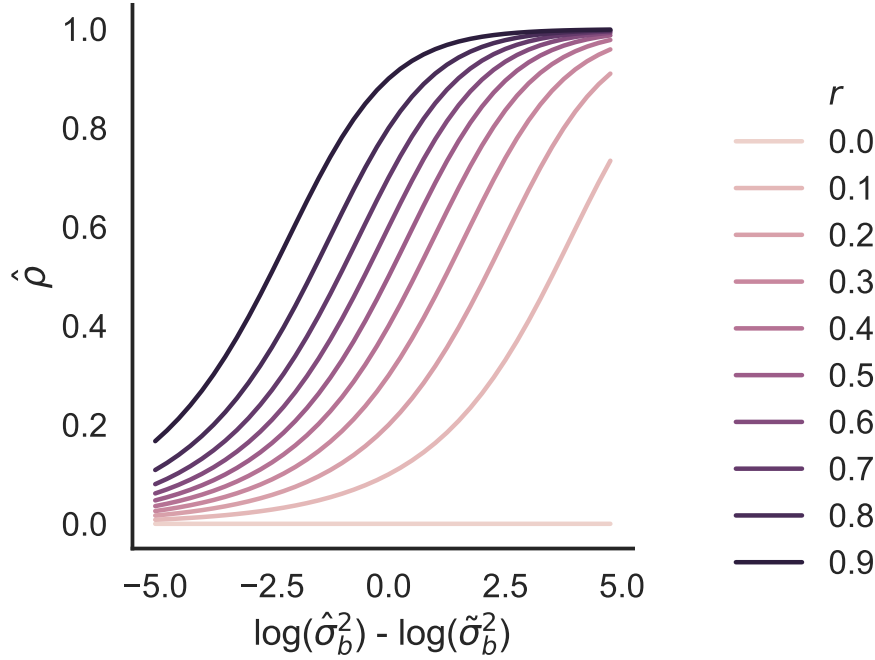


Figure 3.3: Anderson (1957) estimate ($\hat{\rho}$) of the correlation using all of the data as a function of Pearson's r using only n_{nb} complete cases and the difference in the log of the variance of the behavioral data in the n_{nb} complete cases and all n_b cases

$\hat{\rho}$ using all the data will have lower error than estimates for r using only the complete data.

This suggests that there can be cases where, if one variable is more expensive to collect than the other one (for instance neuroimaging data and behavioral data), the same quality of estimates of the correlation can be achieved for lower cost by collecting some data with the more expensive variable missing (Hocking and Smith, 1972).

3.3.2 BEHAVIORAL DATA FOR IMPROVING MODEL ESTIMATES

In the above description of estimation error, we were agnostic as to how exactly parameters were estimated from behavioral data. The most common method of estimating cognitive model parameters for use in a **Model-based Event Related Analysis** is to use maximum

likelihood estimation (Myung, 2003). The maximum likelihood estimate of parameters θ_i for subject i is the set of parameters in a cognitive model C that maximizes the probability of that subject's behavioral data $\mathbf{b}_i$ i.e.

$$\hat{\theta}_i = \operatorname{argmax}_{\theta_i} P(\mathbf{b}_{it} | C, \theta_i, \mathbf{c}_{it}) \quad (3.16)$$

One problem is that many neuroimaging experiments are relatively short so not many trials can be collected per subject. In complex cognitive models, when the number of data points is small, many sets of parameters can be consistent with the data and, therefore, two datasets generated from the cognitive model with the same θ_i could result in very different maximum likelihood estimates $\hat{\theta}_i$, i.e. $\rho(\hat{\theta}_i, \theta_i)$ can be low on average. As demonstrated in the above estimation error section, this will result in lower power for estimating correlations and regression slopes.

Two methods have been used in the literature to increase $\rho(\hat{\theta}_i, \theta_i)$ and $\rho(\hat{\mathbf{c}}_{it}, \mathbf{c}_{it})$: sharing information across subjects or constraining the possible parameter values. One way to use information from other subjects data is to simply assume that all subjects have the same θ values which means that they can be estimated using all of the behavioral data collected in the experiment (Daw, 2011). This increases the data used to fit the parameters by a factor of n . The $\mathbf{c}_{it}$ will still vary by subject because of the dependence on the stimuli in the experiment. If the variability in true parameters θ_i across subjects is small relative to the estimation error, this can be an effective, heuristic way to increase $\rho(\hat{\mathbf{c}}_{it}, \mathbf{c}_{it})$. However, this is not an option in **Individual Differences Analyses** when we are interested the variation across subjects itself. In addition, many cognitive processes do have large individual differences so this may actually decrease $\rho(\hat{\mathbf{c}}_{it}, \mathbf{c}_{it})$.

Another option is to constrain parameter values by adding a prior, $P(\theta_i|\varphi)$. If φ is chosen correctly, using a prior can decrease the likelihood of implausible parameter values such as a learning rate of 0 (Daw, 2011). We can then find the parameters that maximize the posterior, i.e.

$$\hat{\theta}_i = \operatorname{argmax}_{\theta_i} P(\mathbf{b}_{it}|C, \theta_i, \mathbf{c}_{it})P(\theta_i|\varphi) \quad (3.17)$$

This is known as maximum a posteriori estimation (?). By using the prior, we can force θ_i to be closer to more plausible values. However, it can be difficult to know in general if your priors are constraining the parameters to be in the correct space. If your priors are wrong, this could also decrease $\rho(\hat{\theta}_i, \theta_i)$.

One principled way to incorporate information from other subjects and use priors is hierarchical modeling (Gelman, 2006). This is an increasingly popular method in cognitive science and cognitive neuroscience (Ahn et al., 2011; Rouder and Lu, 2005) that assumes that there is a potentially nonuniform distribution $P(\theta_i|\varphi)$ of parameters in the population. Functionally, this constrains the possible values of θ_i as in the prior above but because this is describing the population distribution, we can now estimate φ from all of the subjects data rather than setting φ by hand. The resulting estimate of θ_i will be a weighted average of the estimate from the population with the estimate that just uses an individual's data. The weights are determined by the strength of the individual data. For a subject whose data provides less constraint on the parameters of the cognitive model (i.e. if they had fewer trials or made more inconsistent choices), their estimates will be closer to the prediction from the prior. If a subject made very consistent choices, we might already have a lot of information about the parameters of their cognitive processes and we do not need to use as much information from the population. Thus, hierarchical modeling provides a

way to decide from the data how much we should pool information from the population in creating each individual estimate.

While there exist frequentist methods for estimating hierarchical models (such as restricted maximum likelihood as implemented in popular packages for fitting hierarchical linear models like lmer (Bates et al., 2015)), for arbitrary cognitive models, fitting in this way often requires a complicated derivation or numerical integration which can be challenging in high dimensions. With modern Bayesian inference tools like Stan (Carpenter et al., 2017), it is usually much more straightforward to place a prior on φ and use Bayesian inference to compute a joint posterior over φ and θ . The posterior for the hierarchical model is then

$$P(\theta, \varphi | \mathbf{b}) \propto P(\mathbf{b} | C, \theta, \mathbf{c}) P(\theta_i | \varphi) P(\varphi) \quad (3.18)$$

In addition to Stan, many R or python packages have made hierarchical Bayesian versions of specific cognitive models particularly easy to fit (e.g. drift diffusion models (hddm (Wiecki et al., 2013)) or many types of decision making models (hBayesDM (Ahn et al., 2017))).

The advantage of hierarchical modeling is that it finds an optimal mid-point between pooling all of the information together and fitting a single θ for the entire group vs. estimating a separate θ_i with no information sharing. The individual posterior estimates of θ_i are a weighted combination of the estimate based only on an individual level model and the estimate from the population-level model. The optimal combination is achieved by weighting each individual estimate by the amount of uncertainty in that estimate. Individual estimates that have more uncertainty will result in inferences that are closer to the mean. Katahira (2016) demonstrates that both using individual maximum likelihood estimates and population maximum likelihood estimates are dominated by the hierarchical estimate.

One feature of hierarchical modeling is that individual hierarchical estimates can often be improved simply by sampling more data from the population. This is because the error of the estimates of φ (or the standard deviation of the prior) will, for most models, converge towards the correct estimates with increasing the number of subjects (see simulation in the following section for a demonstration in a simple Gaussian model). This implies that one way to increase $\rho(\hat{\theta}_i, \theta_i)$ or $\rho(\hat{c}_{it}, c_{it})$ for subjects in a neuroimaging experiment is to simply collect behavioral data from more subjects. The rate at which more subjects improve the model will in general depend on the distribution of uncertainty in the individual θ_i as well as the particular form of the model you use.

While not computationally feasible yet for most neuroimaging analyses, joint modeling (Turner et al., 2013) is a recently proposed technique in which the cognitive model and the neuroimaging model are jointly fit, hierarchically, to both the neuroimaging and behavioral data. This can allow for even more efficient use of information by allowing both sources of data to inform the cognitive model parameters. In addition, it allows the uncertainty in the cognitive model parameters to be propagated into the estimates of the regression or correlation, resulting in more powerful estimates without the biases mentioned above. (Turner et al., 2016) demonstrated how joint modeling can be used to improve multimodal inferences even when we don't have data for every mode for every row in the dataset. This can be generalized to the case we discuss where we only have rows in the dataset with only one type of data (e.g. behavior). In this paper we emphasize that an experiment can be designed with extra behavioral data rather than how to combine already collected datasets. In addition, we demonstrate the power of collecting additional data even without using full joint modeling.

In addition, to the statistical methods we mention here, collecting additional behavioral data can be useful in many other ways for improving model estimates. For instance, collecting additional behavioral data in experimental conditions or on novel stimuli not in the neuroimaging experiment could assist in model selection. Choosing the correct model can in practice have a large impact on the correlation between model estimates and true parameters.

3.4 SIMULATIONS

We now construct a simple simulation to show how using the methods described above can improve the precision of inferences about correlations in the data. Following the description above, suppose we have collected a behavioral variable ($\mathbf{b}$) and a neuroimaging variable ($\mathbf{n}$) from n_{nb} subjects. In addition, we collected just behavioral data from n_b subjects. We assume that we have a single behavioral measure and a single neuroimaging measure from each subject. We only observe an estimate of the true behavioral measure $\hat{\mathbf{b}}$, which is estimated from, e.g., a task or questionnaire.

3.4.1 GAUSSIAN MODEL

In order to specify the generative model with missing data, we can create a data matrix for our neuroimaging subjects $\mathbf{bn}$, an n_{nb} -by-2 matrix with $\mathbf{b}_{1:n_{nb}}$ as the first column and $\mathbf{n}$ as the second column. The data matrix is generated according to

$$\mathbf{bn} \sim MVN(\boldsymbol{\mu}^{nb}, \boldsymbol{\Sigma}^{nb}) \quad (3.19)$$

where $\Sigma_{jj}^{nb} = \sigma_j^2$ and $\Sigma_{1,2}^{nb} = \Sigma_{2,1}^{nb} = \rho\sigma_1^2\sigma_2^2$. The additional behavioral subjects then have distribution

$$\mathbf{b}_{(n_{nb}+1):n_b} \sim N(\boldsymbol{\mu}_1^{nb}, \sigma_1^2) \quad (3.20)$$

For the purposes of our simulation, we set $\boldsymbol{\mu}_1^{nb} = 0$ and $\sigma_j^2 = 1$.

We then assume that the behavioral variable is estimated with some noise such that

$$\mathbf{b}_{\text{obs}} \sim N(\mathbf{b}, \sigma_{\mathbf{b}}) \quad (3.21)$$

This could be because a questionnaire has a limited number of questions so the subjects' true score is only known with a limited precision. To keep things simple for estimation purposes, we assume that we have estimated σ_{b_i} separately for each subject. We generate these σ_{b_i} from a uniform distribution between .5 and 1.

Having generated data according to the above model, we now compare several possible methods of inferring the population correlation parameter ρ . Specifically, we compare two ways of estimating $\mathbf{b}$, maximum likelihood and hierarchical posterior mean, and two ways of estimating the correlation, standard Pearson's r and the [Anderson \(1957\)](#) estimate of the correlation using all of the data.

The maximum likelihood estimate of $\mathbf{b}$ is simply $\mathbf{b}_{\text{obs}}$. For the hierarchical model, we need to put a prior on σ_1 . While in practice it is often better to use a weakly or strongly informative prior, for generality, in these simulations, we assume an improper uniform prior, i.e. $P(\sigma_1) \propto 1$. Using the derivation in ([Gelman et al., 2013](#)), we construct a grid approximation to the posterior for $\mathbf{b}$ given $\mathbf{b}_{\text{obs}}$.

We compare estimates with for four values of n_{nb} (25, 50, 100 and 200) and ten values

of ρ from 0 to .9. To investigate the benefit of the additional behavioral data for methods that can make use of the information, for each value of n_{nb} , we test five values of n_b (0, 25, 50, 100, 200). Finally, to compare the benefit of the additional behavioral data to collecting an additional neuroimaging subject, we compare all estimates to Pearson's r with one additional neuroimaging subject, both using maximum likelihood and hierarchical estimation for the behavioral data. For each set of $(n_n b, n_b, \rho)$, we generate 10,000 datasets.

Of note here is that collecting fMRI data can often cost in the range of \$450 (?) per hour of subject time. In addition to the price, collecting neuroimaging data can have many other nonmonetary costs such as requiring the subject to come into the lab, requiring the time of trained researchers or technicians to run the machine and only being able to run one subject at a time. In contrast, many behavioral studies can be easily run in parallel online without significant monitoring by researchers for a cost of around \$11 per hour of subject time (Hara et al., 2018; Crump et al., 2013). Thus, if 40 behavioral subjects can improve inferences more than a single fMRI subject, this might be a worthwhile tradeoff in many experimental designs.

In Figure 3.5 we can see how additional behavioral data can influence the mean squared error of the correlation estimate using all of our methods of estimation. We can see that hierarchical estimation always improves estimates of the correlation over the maximum likelihood estimate, as was found by Katahira (Katahira, 2016). However, adding additional behavioral data to the hierarchical model only improved estimates for the correlation using Pearson's r slightly. Comparing the two methods of correlation estimation, we find that the effect of the additional behavioral data can be nonlinear in the size of the underlying neuroimaging dataset and the true value of ρ . Having too few neuroimaging data points

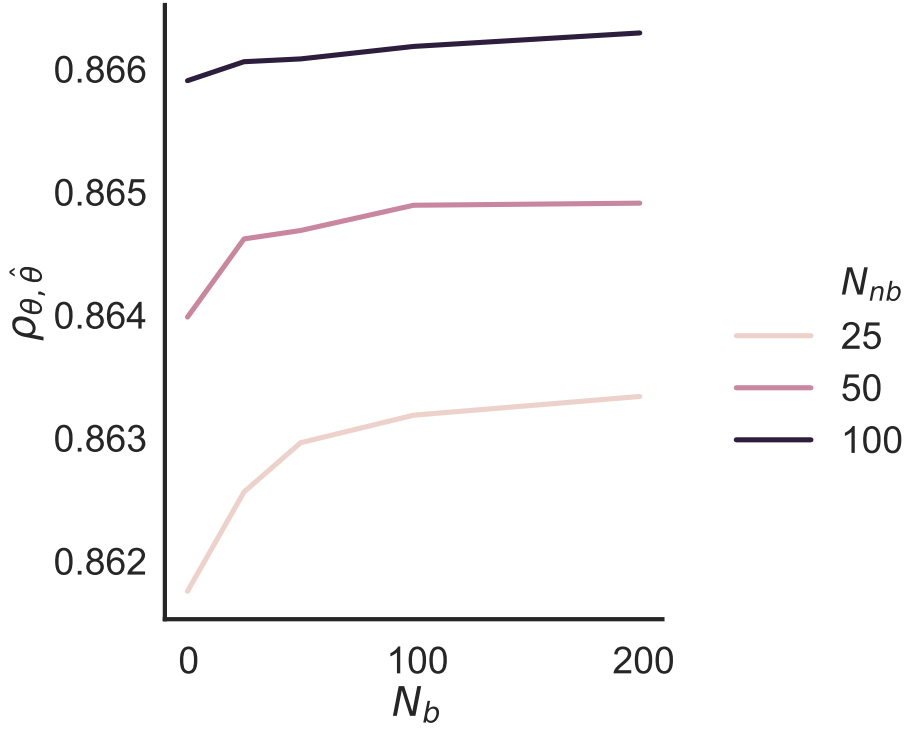


Figure 3.4: Correlation between the estimated $\hat{\theta}$ and θ from the hierarchical model, averaged over 10,000 simulated datasets, as a function of the number of neuroimaging subjects n_{nb} and number of additional behavioral subjects n_b .

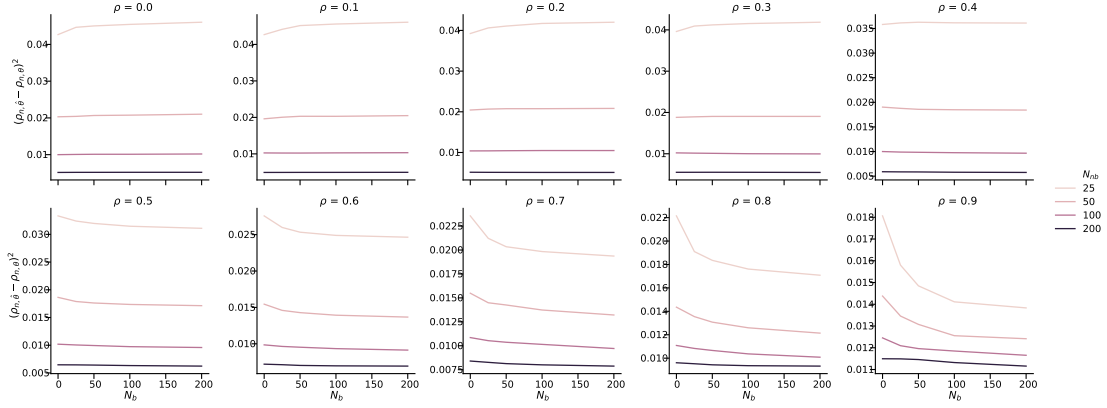


Figure 3.5: Mean squared estimation error, averaged over 10,000 simulated datasets, for the estimate of ρ from the four correlation estimation methods as a function of the true correlation, the number of neuroimaging subjects n_{nb} and number of additional behavioral subjects n_b . We also compare to estimates from a neuroimaging dataset with $n_{nb} + 1$ subjects

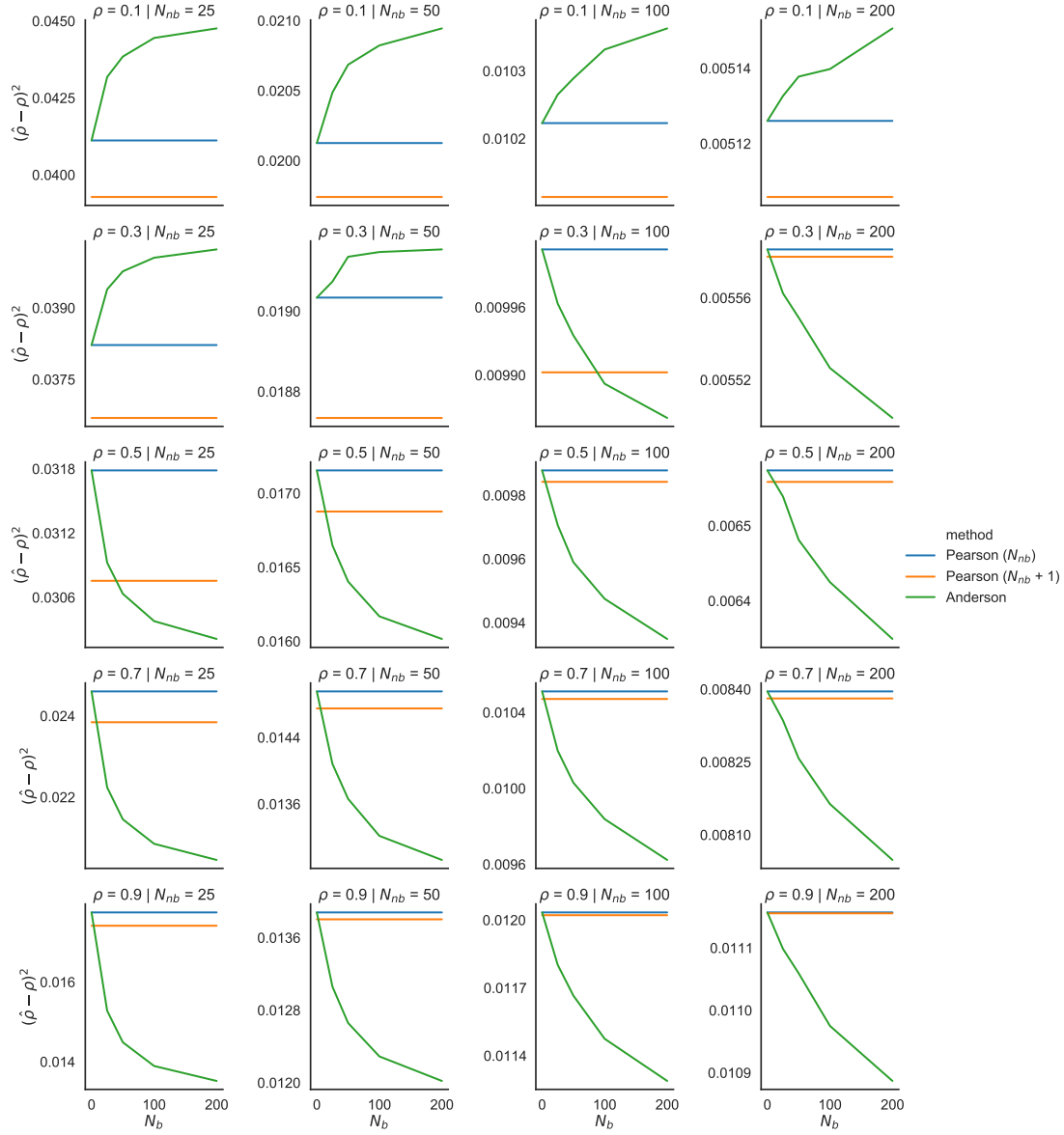


Figure 3.6: Mean squared estimation error, averaged over 10,000 simulated datasets, for the estimate of ρ from the four correlation estimation methods as a function of the true correlation, the number of neuroimaging subjects $n_n b$ and number of additional behavioral subjects n_b . We also compare to estimates from a neuroimaging dataset with $n_n b + 1$ subjects

to estimate a weak correlation (e.g. only 25 subjects for a correlation of .1) makes the behavioral data less useful and it can be better to collect more neuroimaging data. This finding resembles similar results from (Garren, 1998). At low amounts of neuroimaging data and lower true values of ρ , the Anderson estimate of the correlation using all of the behavioral data can actually perform worse than the standard Pearson estimate. However, as the true ρ and number of neuroimaging subjects increase, the value of the additional data increases such that collecting additional behavioral data can be better than collecting a smaller amount of additional neuroimaging data. In some cases, collecting just 25 behavioral subjects can improve precision more than another neuroimaging subject. At the cost levels described above, this can mean that behavioral data can be a better investment than equivalently priced neuroimaging data. Therefore, design analysis taking costs into account can recommend collecting behavioral data rather than fMRI data!

The mean squared error can be decomposed into the bias and the variance, i.e.

$$MSE(\hat{\rho}, \rho) = E((\hat{\rho} - \rho)^2) = Bias(\hat{\rho}, \rho)^2 + Var(\hat{\rho}) \quad (3.22)$$

where Bias is defined as

$$Bias(\hat{\rho}, \rho) = E(\hat{\rho} - \rho) \quad (3.23)$$

As with mean squared error, we can also look at how various methods perform in terms of bias, i.e. is the estimate of ρ equal to ρ on average? It is well known that Pearson's r is biased downward such that, on average, correlation estimates will be lower than their true value (Olkin and Pratt, 1958). In Figure 3.7, we can see that both hierarchical estimates of $\mathbf{b}$ and the Anderson estimate of ρ decrease the bias in estimating ρ relative to their maximum like-

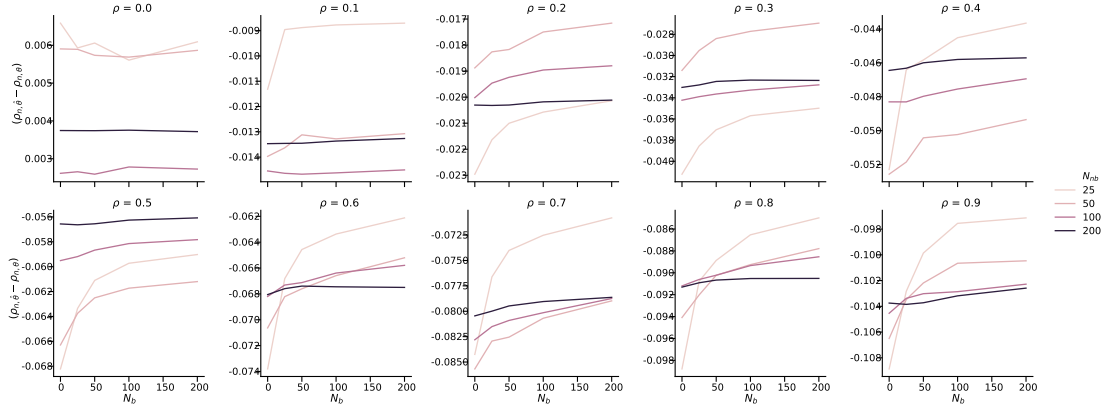


Figure 3.7: Bias in estimate of ρ , over 10,000 simulated datasets, for the four estimation methods as a function of as a function of the true correlation, the number of neuroimaging subjects n_{nb} and number of additional behavioral subjects n_b . We also compare to estimates from a neuroimaging dataset with $n_{nb}+1$ subjects

likelihood and Pearson counterparts. This means that the increase in mean squared error in the Anderson estimate for lower values of ρ and n_{nb} is due to the behavioral data adding variance to the estimate. For some research questions and statistical inferences, we may care more about bias or variance of the estimator.

3.5 COLLECTING ADDITIONAL BEHAVIORAL DATA

We are not the first to understand the power of collecting additional behavioral data in interpreting neuroimaging signals. As a means of clarifying our point we describe below several examples of how this principle has already been put into practice by cognitive neuroscientists. We then demonstrate through simulation how joint modeling in particular could allow for the most efficient use of behavioral data and provide cognitive neuroscientists with a means of extracting the most information out of their data.

The idea of collecting additional behavioral data outside of your main experimental group in order to reduce bias and variance in cognitive parameters has a long history in

psychology and cognitive neuroscience. In computational cognitive neuroscience we are often dealing with a) latent stimulus properties that are cognitively relevant (e.g. valence) or b) latent parameters of the cognitive process that cannot be directly related to behavior (e.g. learning rates). Without large amounts of data to precisely infer the parameters, relating (functions of) them to the brain is very difficult for the reasons discussed above. We discuss below several ways in which computational cognitive neuroscientists have tried to deal with these two issues.

3.5.1 STIMULI NORMING

The range of possible stimuli in many psychology studies is vast and many models of cognition are specified in terms that cannot be observed directly from the stimulus itself. For instance, in many models of language, the concreteness or emotional valence of a word might affect production (?). While it would be ideal to collect this data directly from the subjects in the main experiment themselves, this might be too demanding of subjects time and cognitive effort to collect in a single experimental session. In addition, individual subjects might be fairly noisy in their responses and how they interpret the ratings so collecting additional data can help smooth out these measurements. Therefore, a common practice is to collect a set of stimulus "norms" derived from a large survey experiment done in the population of interest (or at least a similar population). Some function of these ratings (often the mean or the mean for some subgroup) are used effectively as properties of the stimulus and can then be plugged in to any cognitive or statistical model. This approach goes back to at least to the Kent and Rosanoff norms for word association (Kent and Rosanoff, 1910).

More recently this approach has been used in cognitive neuroscience as well. In order

to study semantic representations of words in the brain, Bruffaerts (Bruffaerts et al., 2013) collected feature property lists for 196 animals from over a thousand behavioral subjects and feature applicability judgments from a separate set of 764 participants. The cosine similarity computed from the applicability judgments was then used as a representation that could be related to fMRI measurements. In this case, collecting data outside the scanner allowed for longer lists of possible features and judgments that were unbiased by collecting the previous task. In addition, no one subject had to answer every question for every stimulus and aggregating the judgments over the group allowed the experimenters to significantly reduce the noise in their cognitive variable (the semantic similarity between two words). Similarly, in a recent study, Luke Chang and colleagues collected fMRI data of several subjects watching episodes of a television show with the goal of understanding the role of vmPFC in processing emotions. Asking subjects while in the scanner to rate their emotions (and asking too frequently) might have qualitatively changed thought processes. Instead, Chang and colleagues collected a large dataset of online subjects rating their experience of various emotions at different points during the video. By collecting a large dataset, they were able to only interrupt individual subjects relatively infrequently so as to not alter the viewing process too much and then used machine learning to estimate an aggregate timecourse of emotional experience. This allowed them to find evidence that vmPFC had at least two states that corresponded to positive and negative affect respectively.

3.5.2 MODEL-BASED fMRI

Cognitive scientists have long used computational models as a way of understanding the latent processes involved in behavior. In the mid-2000s, many cognitive neuroscientists

realized that using these computational models might be the only way of discovering the functions of many brain regions that were involved in non-sensory cognitive processes such as decision-making. The typical procedure for these studies is to use each subject's behavior while from their scanner session to obtain point estimates of cognitive model parameters. These estimates or some function of the estimates and the stimuli (e.g. the reward prediction error in popular reinforcement learning models) are then used as a cognitive variable that can be related to the fMRI signal. However, using a maximum likelihood model fit to behavioral data collected in a single fMRI session, while perhaps an unbiased estimate of cognitive parameters, can be very noisy. Because measurement error in the cognitive variable can attenuate and add noise to estimates of the correlation with BOLD signal, this can be a poor use of behavioral data. In a paper on best practices for conducting model-based fMRI analyses, Daw mentioned: One practical point is that, in our experience (Daw et al., 2006; Schonberg et al., 2007), multisubject fMRI results from analyses of this sort are, in practice, more robust if a single α and a single set of any other parameters of the learning model is used to generate regressors for all subjects. (Daw, 2011) While Daw does not suggest collecting data outside of the fMRI study, this can be thought of as a simple way of incorporating more behavioral data into the estimates of the cognitive model parameters so as to decrease measurement error in the cognitive variable.

3.5.3 HIERARCHICAL MODEL-BASED fMRI

Rather than just using a single set of parameters fit to the entire dataset, one way to flexibly take into account individual differences in cognitive processes while still allowing the model to share information across subjects is through hierarchical modeling. (Ahn et al., 2011)

demonstrated that hierarchical modeling can improve parameter estimation in commonly used decision making models and that this can lead to both improved fits to the neural signal as well as qualitatively different maps of the neural signal. Since then, the use of hierarchical fitting for cognitive models has become much more common in model-based fMRI. However, despite it being clear that collecting behavioral data outside the scanner can improve the fit of the hierarchical model (especially when trying to capture group differences such as with patient populations), we are unaware of a study that collects behavioral data beyond the set of participants run in the scanner.

3.6 CONCLUSION

While we have mostly discussed collecting additional behavioral data in the context of designing a new study, it could also be a consideration in a study that resuses an old dataset. It could be that a model is not identifiable with a particular dataset but it could become identifiable by collecting behavioral data in more conditions. With the advent of more open fMRI data, joint modeling could allow modelers to learn new things from old datasets simply by collecting more cheap behavioral data.

4

Conclusion

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Morbi commodo, ipsum sed pharetra gravida, orci magna rhoncus neque, id pulvinar odio lorem non turpis. Nullam sit amet enim. Suspendisse id velit vitae ligula volutpat condimentum. Aliquam erat volutpat. Sed quis velit. Nulla facilisi. Nulla libero. Vivamus pharetra posuere sapien. Nam consectetur. Sed aliquam, nunc eget euismod ullamcorper, lectus nunc ullamcorper orci,

fermentum bibendum enim nibh eget ipsum. Donec porttitor ligula eu dolor. Maecenas vitae nulla consequat libero cursus venenatis. Nam magna enim, accumsan eu, blandit sed, blandit a, eros.

Quisque facilisis erat a dui. Nam malesuada ornare dolor. Cras gravida, diam sit amet rhoncus ornare, erat elit consectetur erat, id egestas pede nibh eget odio. Proin tincidunt, velit vel porta elementum, magna diam molestie sapien, non aliquet massa pede eu diam. Aliquam iaculis. Fusce et ipsum et nulla tristique facilisis. Donec eget sem sit amet ligula viverra gravida. Etiam vehicula urna vel turpis. Suspendisse sagittis ante a urna. Morbi a est quis orci consequat rutrum. Nullam egestas feugiat felis. Integer adipiscing semper ligula. Nunc molestie, nisl sit amet cursus convallis, sapien lectus pretium metus, vitae pretium enim wisi id lectus. Donec vestibulum. Etiam vel nibh. Nulla facilisi. Mauris pharetra. Donec augue. Fusce ultrices, neque id dignissim ultrices, tellus mauris dictum elit, vel lacinia enim metus eu nunc.

Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Vestibulum tortor quam, feugiat vitae, ultricies eget, tempor sit amet, ante. Donec eu libero sit amet quam egestas semper. Aenean ultricies mi vitae est. Mauris placerat eleifend leo. Quisque sit amet est et sapien ullamcorper pharetra. Vestibulum erat wisi, condimentum sed, commodo vitae, ornare sit amet, wisi. Aenean fermentum, elit eget tincidunt condimentum, eros ipsum rutrum orci, sagittis tempus lacus enim ac dui. Donec non enim in turpis pulvinar facilisis. Ut felis.

Cras sed ante. Phasellus in massa. Curabitur dolor eros, gravida et, hendrerit ac, cursus non, massa. Aliquam lorem. In hac habitasse platea dictumst. Cras eu mauris. Quisque lacus. Donec ipsum. Nullam vitae sem at nunc pharetra ultricies. Vivamus elit eros, ullam-

corper a, adipiscing sit amet, porttitor ut, nibh. Maecenas adipiscing mollis massa. Nunc ut dui eget nulla venenatis aliquet. Sed luctus posuere justo. Cras vehicula varius turpis. Vivamus eros metus, tristique sit amet, molestie dignissim, malesuada et, urna.

Cras dictum. Maecenas ut turpis. In vitae erat ac orci dignissim eleifend. Nunc quis justo. Sed vel ipsum in purus tincidunt pharetra. Sed pulvinar, felis id consectetur malesuada, enim nisl mattis elit, a facilisis tortor nibh quis leo. Sed augue lacus, pretium vitae, molestie eget, rhoncus quis, elit. Donec in augue. Fusce orci wisi, ornare id, mollis vel, lacinia vel, massa.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Morbi commodo, ipsum sed pharetra gravida, orci magna rhoncus neque, id pulvinar odio lorem non turpis. Nullam sit amet enim. Suspendisse id velit vitae ligula volutpat condimentum. Aliquam erat volutpat. Sed quis velit. Nulla facilisi. Nulla libero. Vivamus pharetra posuere sapien. Nam consectetur. Sed aliquam, nunc eget euismod ullamcorper, lectus nunc ullamcorper orci, fermentum bibendum enim nibh eget ipsum. Donec porttitor ligula eu dolor. Maecenas vitae nulla consequat libero cursus venenatis. Nam magna enim, accumsan eu, blandit sed, blandit a, eros.

Quisque facilisis erat a dui. Nam malesuada ornare dolor. Cras gravida, diam sit amet rhoncus ornare, erat elit consectetur erat, id egestas pede nibh eget odio. Proin tincidunt, velit vel porta elementum, magna diam molestie sapien, non aliquet massa pede eu diam. Aliquam iaculis. Fusce et ipsum et nulla tristique facilisis. Donec eget sem sit amet ligula viverra gravida. Etiam vehicula urna vel turpis. Suspendisse sagittis ante a urna. Morbi a est quis orci consequat rutrum. Nullam egestas feugiat felis. Integer adipiscing semper ligula. Nunc molestie, nisl sit amet cursus convallis, sapien lectus pretium metus, vitae pretium

enim wisi id lectus. Donec vestibulum. Etiam vel nibh. Nulla facilisi. Mauris pharetra.
Donec augue. Fusce ultrices, neque id dignissim ultrices, tellus mauris dictum elit, vel
lacinia enim metus eu nunc.



Classifier Plots

Todo

References

- Ahn, W.-Y., Haines, N., and Zhang, L. (2017). Revealing Neurocomputational Mechanisms of Reinforcement Learning and Decision-Making With the hBayesDM Package. *Computational Psychiatry*, 1:24–57.
- Ahn, W.-Y., Krawitz, A., Kim, W., Busemeyer, J. R., and Brown, J. W. (2011). A model-based fMRI analysis with hierarchical Bayesian parameter estimation. *Journal of Neuroscience, Psychology, and Economics*, 4(2):95–110.
- Anderson, J. R. (2012). Tracking problem solving by multivariate pattern analysis and Hidden Markov Model algorithms. *Neuropsychologia*, 50(4):487–498.
- Anderson, J. R., Betts, S., Ferris, J. L., and Fincham, J. M. (2010). Neural imaging to track mental states while using an intelligent tutoring system. *Proceedings of the National Academy of Sciences*, 107(15):7018–7023.
- Anderson, J. R., Corbett, A. T., Koedinger, K. R., and Pelletier, R. (1995). Cognitive Tutors: Lessons Learned. *Journal of the Learning Sciences*, 4(2):167–207.
- Anderson, J. R., Fincham, J. M., Schneider, D. W., and Yang, J. (2012). Using brain imaging to track problem solving in a complex state space. *NeuroImage*, 60(1):633–643.
- Anderson, T. W. (1957). Maximum Likelihood Estimates for a Multivariate Normal Distribution when Some Observations are Missing. *Journal of the American Statistical Association*, 52(278):200–203.
- Atkinson, R. C. (1972). Optimizing the learning of a second-language vocabulary. *Journal of Experimental Psychology*, 96(1):124–129.
- Avants, B. B., Tustison, N. J., Song, G., Cook, P. A., Klein, A., and Gee, J. C. (2011). A reproducible evaluation of ANTs similarity metric performance in brain image registration. *NeuroImage*, 54(3):2033–2044.

- Bainbridge, W. A. (2019). Memorability: How what we see influences what we remember. In *Psychology of Learning and Motivation*, volume 70, pages 1–27. Elsevier.
- Bainbridge, W. A., Dilks, D. D., and Oliva, A. (2017). Memorability: A stimulus-driven perceptual neural signature distinctive from memory. *NeuroImage*, 149:141–152.
- Bainbridge, W. A. and Rissman, J. (2018). Dissociating neural markers of stimulus memorability and subjective recognition during episodic retrieval. *Scientific Reports*, 8(1):8679.
- Baker, R. S. J. d., Corbett, A. T., and Aleven, V. (2008). More Accurate Student Modeling through Contextual Estimation of Slip and Guess Probabilities in Bayesian Knowledge Tracing. In Woolf, B. P., Aïmeur, E., Nkambou, R., and Lajoie, S., editors, *Intelligent Tutoring Systems*, pages 406–415, Berlin, Heidelberg. Springer Berlin Heidelberg.
- Baker, R. S. J. D., Goldstein, A. B., and Heffernan, N. T. (2011). Detecting Learning Moment-by-Moment. *Int. J. Artif. Intell. Ed.*, 21(1–2):5–25. Place: NLD Publisher: IOS Press.
- Bates, D., Mächler, M., Bolker, B., and Walker, S. (2015). Fitting Linear Mixed-Effects Models Using **lme4**. *Journal of Statistical Software*, 67(1).
- Behzadi, Y., Restom, K., Liau, J., and Liu, T. T. (2007). A component based noise correction method (CompCor) for BOLD and perfusion based fMRI. *NeuroImage*, 37(1):90–101.
- Benjamini, Y. and Hochberg, Y. (1995). Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300.
- Benjamini, Y. and Yekutieli, D. (2001). THE CONTROL OF THE FALSE DISCOVERY RATE IN MULTIPLE TESTING UNDER DEPENDENCY. *The Annals of Statistics*, 29(4):1165–1188.
- Bondareva, D., Conati, C., Feyzi-Behnagh, R., Harley, J. M., Azevedo, R., and Bouchet, F. (2013). Inferring Learning from Gaze Data during Interaction with an Environment to Support Self-Regulated Learning. In Lane, H. C., Yacef, K., Mostow, J., and Pavlik, P., editors, *Artificial Intelligence in Education*, pages 229–238, Berlin, Heidelberg. Springer Berlin Heidelberg.
- Brown, E. N. and Behrmann, M. (2017). Controversy in statistical analysis of functional magnetic resonance imaging data. *Proceedings of the National Academy of Sciences*, 114(17):E3368–E3369.

- Bruett, H., Calloway, R. C., Tokowicz, N., and Coutanche, M. N. (2020). Neural pattern similarity across concept exemplars predicts memory after a long delay. *NeuroImage*, 219:117030.
- Bruffaerts, R., Dupont, P., Peeters, R., De Deyne, S., Storms, G., and Vandenberghe, R. (2013). Similarity of fMRI Activity Patterns in Left Perirhinal Cortex Reflects Semantic Similarity between Words. *Journal of Neuroscience*, 33(47):18597–18607.
- Brunec, I. K., Robin, J., Olsen, R. K., Moscovitch, M., and Barense, M. D. (2020). Integration and differentiation of hippocampal memory traces. *Neuroscience & Biobehavioral Reviews*, page S0149763420304838.
- Calhoun, V., Adali, T., Pearlson, G., and Pekar, J. (2001). A method for making group inferences from functional MRI data using independent component analysis. *Human Brain Mapping*, 14(3):140–151.
- Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M., Guo, J., Li, P., and Riddell, A. (2017). *Stan* : A Probabilistic Programming Language. *Journal of Statistical Software*, 76(1).
- Cessie, S. L. and Houwelingen, J. C. V. (1992). Ridge Estimators in Logistic Regression. *Applied Statistics*, 41(1):191.
- Chadwick, M. J., Anjum, R. S., Kumaran, D., Schacter, D. L., Spiers, H. J., and Hassabis, D. (2016). Semantic representations in the temporal pole predict false memories. *Proceedings of the National Academy of Sciences*, 113(36):10180–10185.
- Charest, I., Kievit, R. A., Schmitz, T. W., Deca, D., and Kriegeskorte, N. (2014). Unique semantic space in the brain of each beholder predicts perceived similarity. *Proceedings of the National Academy of Sciences*, 111(40):14565–14570.
- Chen, J., Leong, Y. C., Honey, C. J., Yong, C. H., Norman, K. A., and Hasson, U. (2017). Shared memories reveal shared structure in neural activity across individuals. *Nature Neuroscience*, 20(1):115–125.
- Chen, L., Wassermann, D., Abrams, D. A., Kochalka, J., Gallardo-Diez, G., and Menon, V. (2019). The visual word form area (VWFA) is part of both language and attention circuitry. *Nature Communications*, 10(1):5601.
- Chikazoe, J., Lee, D. H., Kriegeskorte, N., and Anderson, A. K. (2014). Population coding of affect across stimuli, modalities and individuals. *Nature Neuroscience*, 17(8):1114–1122.

- Corbett, A. T. and Anderson, J. R. (1995). Knowledge tracing: Modeling the acquisition of procedural knowledge. *User Modelling and User-Adapted Interaction*, 4(4):253–278.
- Cox, H. (1997). Software tools for analysis and visualization of fMRI data. *NMR IN BIOMEDICINE*, 10:8.
- Craik, F. I. M. and Tulving, E. (1975). Depth of processing and the retention of words in episodic memory. *Journal of Experimental Psychology: General*, 104(3):268–294.
- Cremers, H. R., Wager, T. D., and Yarkoni, T. (2017). The relation between statistical power and inference in fMRI. *PLOS ONE*, 12(11):e0184923.
- Crump, M. J. C., McDonnell, J. V., and Gureckis, T. M. (2013). Evaluating Amazon’s Mechanical Turk as a Tool for Experimental Behavioral Research. *PLoS ONE*, 8(3):e57410.
- Dale, A. M., Fischl, B., and Sereno, M. I. (1999). Cortical Surface-Based Analysis. *NeuroImage*, 9(2):179–194.
- Danker, J. F., Tompary, A., and Davachi, L. (2016). Trial-by-Trial Hippocampal Encoding Activation Predicts the Fidelity of Cortical Reinstatement During Subsequent Retrieval. *Cerebral Cortex*, page bhw146.
- Davachi, L. (2006). Item, context and relational episodic encoding in humans. *Current Opinion in Neurobiology*, 16(6):693–700.
- Davachi, L., Mitchell, J. P., and Wagner, A. D. (2003). Multiple routes to memory: Distinct medial temporal lobe processes build item and source memories. *Proceedings of the National Academy of Sciences*, 100(4):2157–2162.
- Davis, T., Xue, G., Love, B. C., Preston, A. R., and Poldrack, R. A. (2014). Global Neural Pattern Similarity as a Common Basis for Categorization and Recognition Memory. *Journal of Neuroscience*, 34(22):7472–7484.
- Daw, N. D. (2011). Trial-by-trial data analysis using computational models: (Tutorial Review). In *Decision Making, Affect, and Learning*. Oxford University Press, Oxford.
- Dawes, R. M. (1979). The robust beauty of improper linear models in decision making. *American Psychologist*, 34(7):571–582.
- DeLosh, E. L. and McDaniel, M. A. (1996). The role of order information in free recall: Application to the word-frequency effect. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 22(5):1136–1146.

Dietterich, T. G. (1998). Approximate Statistical Tests for Comparing Supervised Classification Learning Algorithms. *Neural Computation*, 10(7):1895–1923.

Dikker, S., Wan, L., Davidesco, I., Kaggen, L., Oostrik, M., McClintock, J., Rowland, J., Michalareas, G., Van Bavel, J. J., Ding, M., and Poeppel, D. (2017). Brain-to-Brain Synchrony Tracks Real-World Dynamic Group Interactions in the Classroom. *Current Biology*, 27(9):1375–1380.

Doroudi, S. and Brunskill, E. (2017). The Misidentified Identifiability Problem of Bayesian Knowledge Tracing. *International Educational Data Mining Society*. Publisher: ERIC.

Durnez, J., Blair, R., and Poldrack, R. A. (2017). Neurodesign: Optimal Experimental Designs for Task fMRI. preprint, Neuroscience.

Esteban, O., Markiewicz, C. J., Blair, R. W., Moodie, C. A., Isik, A. I., Erramuzpe, A., Kent, J. D., Goncalves, M., DuPre, E., Snyder, M., Oya, H., Ghosh, S. S., Wright, J., Durnez, J., Poldrack, R. A., and Gorgolewski, K. J. (2019). fMRIPrep: a robust pre-processing pipeline for functional MRI. *Nature Methods*, 16(1):111–116.

Feinberg, D. A. and Yacoub, E. (2012). The rapid development of high speed, resolution and precision in fMRI. *NeuroImage*, 62(2):720–725.

Feng, J. (2017). Essays on Learning Through Practice. page 140.

Feng, K., Zhao, X., Liu, J., Cai, Y., Ye, Z., Chen, C., and Xue, G. (2019). Spaced Learning Enhances Episodic Memory by Increasing Neural Pattern Similarity Across Repetitions. *The Journal of Neuroscience*, 39(27):5351–5360.

Fisher, R. A. (1915). Frequency Distribution of the Values of the Correlation Coefficient in Samples from an Indefinitely Large Population. *Biometrika*, 10(4):507.

Fonov, V., Evans, A. C., Botteron, K., Almli, C. R., McKinstry, R. C., and Collins, D. L. (2011). Unbiased average age-appropriate atlases for pediatric studies. *NeuroImage*, 54(1):313–327.

Friedman, J., Hastie, T., and Tibshirani, R. (2010). Regularization Paths for Generalized Linear Models via Coordinate Descent. *Journal of Statistical Software*, 33(1).

Friston, K. J., Holmes, A. P., Worsley, K. J., Poline, J.-P., Frith, C. D., and Frackowiak, R. S. J. (1994). Statistical parametric maps in functional imaging: A general linear approach. *Human Brain Mapping*, 2(4):189–210.

- Fukuda, K. and Woodman, G. F. (2015). Predicting and Improving Recognition Memory Using Multiple Electrophysiological Signals in Real Time. *Psychological Science*, 26(7):1026–1037.
- Garren, S. T. (1998). Maximum likelihood estimation of the correlation coefficient in a bivariate normal model with missing data. *Statistics & Probability Letters*, 38(3):281–288.
- Gelman, A. (2006). Multilevel (Hierarchical) Modeling: What It Can and Cannot Do. *Technometrics*, 48(3):432–435.
- Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2013). *Bayesian Data Analysis*. Chapman and Hall/CRC, 0 edition.
- Gelman, A. and Rubin, D. B. (1992). Inference from Iterative Simulation Using Multiple Sequences. *Statistical Science*, 7(4):457–472.
- Gelman, A. and Tuerlinckx, F. (2000). Type S error rates for classical and Bayesian single and multiple comparison procedures. *Computational Statistics*, 15(3):373–390.
- Gillund, G. and Shiffrin, R. M. (1984). A retrieval model for both recognition and recall. *Psychological Review*, 91(1):1–67.
- Golland, P. and Fischl, B. (2003). Permutation Tests for Classification: Towards Statistical Significance in Image-Based Studies. In Goos, G., Hartmanis, J., van Leeuwen, J., Taylor, C., and Noble, J. A., editors, *Information Processing in Medical Imaging*, volume 2732, pages 330–341. Springer Berlin Heidelberg, Berlin, Heidelberg. Series Title: Lecture Notes in Computer Science.
- Gorgolewski, K., Burns, C. D., Madison, C., Clark, D., Halchenko, Y. O., Waskom, M. L., and Ghosh, S. S. (2011). Nipype: A Flexible, Lightweight and Extensible Neuroimaging Data Processing Framework in Python. *Frontiers in Neuroinformatics*, 5.
- Greve, D. N. and Fischl, B. (2009). Accurate and robust brain image alignment using boundary-based registration. *NeuroImage*, 48(1):63–72.
- Grill-Spector, K., Henson, R., and Martin, A. (2006). Repetition and the brain: neural models of stimulus-specific effects. *Trends in Cognitive Sciences*, 10(1):14–23.
- Grimaldi, P. J., Pyc, M. A., and Rawson, K. A. (2010). Normative multitrial recall performance, metacognitive judgments, and retrieval latencies for Lithuanian—English paired associates. *Behavior Research Methods*, 42(3):634–642.

- Hara, K., Adams, A., Milland, K., Savage, S., Callison-Burch, C., and Bigham, J. P. (2018). A Data-Driven Analysis of Workers' Earnings on Amazon Mechanical Turk. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems - CHI '18*, pages 1–14, Montreal QC, Canada. ACM Press.
- Hasson, U., Furman, O., Clark, D., Dudai, Y., and Davachi, L. (2008). Enhanced Inter-subject Correlations during Movie Viewing Correlate with Successful Episodic Encoding. *Neuron*, 57(3):452–462.
- Haxby, J., Guntupalli, J., Connolly, A., Halchenko, Y., Conroy, B., Gobbini, M., Hanke, M., and Ramadge, P. (2011). A Common, High-Dimensional Model of the Representational Space in Human Ventral Temporal Cortex. *Neuron*, 72(2):404–416.
- Haxby, J. V., Guntupalli, J. S., Nastase, S. A., and Feilong, M. (2020). Hyperalignment: Modeling shared information encoded in idiosyncratic cortical topographies. *eLife*, 9:e56601.
- Hintzman, D. L. (1988). Judgments of frequency and recognition memory in a multiple-trace memory model. *Psychological Review*, 95(4):528–551.
- Hocking, R. R. and Smith, W. B. (1972). Optimum Incomplete Multinormal Samples. *Technometrics*, 14(2):299–307.
- Hoerl, A. E. and Kennard, R. W. (1970). Ridge Regression: Biased Estimation for Nonorthogonal Problems. *Technometrics*, 12(1):55–67.
- Iordan, M. C., Greene, M. R., Beck, D. M., and Fei-Fei, L. (2016). Typicality sharpens category representations in object-selective cortex. *NeuroImage*, 134:170–179.
- Jenkinson, M., Bannister, P., Brady, M., and Smith, S. (2002). Improved Optimization for the Robust and Accurate Linear Registration and Motion Correction of Brain Images. *NeuroImage*, 17(2):825–841.
- Jenkinson, M. and Smith, S. (2001). A global optimisation method for robust affine registration of brain images. *Medical Image Analysis*, 5(2):143–156.
- Kahana, M. J., Aggarwal, E. V., and Phan, T. D. (2018). The variability puzzle in human memory. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 44(12):1857–1863.
- Kanwisher, N., McDermott, J., and Chun, M. M. (1997). The Fusiform Face Area: A Module in Human Extrastriate Cortex Specialized for Face Perception. *The Journal of Neuroscience*, 17(11):4302–4311.

- Katahira, K. (2016). How hierarchical models improve point estimates of model parameters at the individual level. *Journal of Mathematical Psychology*, 73:37–58.
- Kent, G. H. and Rosanoff, A. J. (1910). Part II. Association in insane subjects. In *A study of association in insanity*, pages 16–72. American Journal of Insanity.
- Kim, H. (2011). Neural activity that predicts subsequent memory and forgetting: A meta-analysis of 74 fMRI studies. *NeuroImage*, 54(3):2446–2461.
- Klein, A., Ghosh, S. S., Bao, F. S., Giard, J., Häme, Y., Stavsky, E., Lee, N., Rossa, B., Reuter, M., Chaibub Neto, E., and Keshavan, A. (2017). Mindboggling morphometry of human brains. *PLOS Computational Biology*, 13(2):e1005350.
- Knutson, B., Westdorp, A., Kaiser, E., and Hommer, D. (2000). FMRI Visualization of Brain Activity during a Monetary Incentive Delay Task. *NeuroImage*, 12(1):20–27.
- Koch, G. E., Paulus, J. P., and Coutanche, M. N. (2020). Neural Patterns are More Similar across Individuals during Successful Memory Encoding than during Failed Memory Encoding. *Cerebral Cortex*, 30(7):3872–3883.
- Koedinger, K. R., Brunskill, E., Baker, R. S., McLaughlin, E. A., and Stamper, J. (2013). New Potentials for Data-Driven Intelligent Tutoring System Development and Optimization. *AI Magazine*, 34(3):27.
- Krakauer, J. W., Ghazanfar, A. A., Gomez-Marin, A., MacIver, M. A., and Poeppel, D. (2017). Neuroscience Needs Behavior: Correcting a Reductionist Bias. *Neuron*, 93(3):480–490.
- Kuhl, B. A., Rissman, J., and Wagner, A. D. (2012). Multi-voxel patterns of visual category representation during episodic encoding are predictive of subsequent memory. *Neuropsychologia*, 50(4):458–469.
- Kuhn, M. and Wickham, H. (2020). *Tidymodels: a collection of packages for modeling and machine learning using tidyverse principles*.
- Lanczos, C. (1964). A Precision Approximation of the Gamma Function. *Journal of the Society for Industrial and Applied Mathematics Series B Numerical Analysis*, 1(1):86–96.
- LaRocque, K. F., Smith, M. E., Carr, V. A., Witthoft, N., Grill-Spector, K., and Wagner, A. D. (2013). Global Similarity and Pattern Separation in the Human Medial Temporal Lobe Predict Subsequent Memory. *Journal of Neuroscience*, 33(13):5466–5474.

- Lindquist, M. A., Meng Loh, J., Atlas, L. Y., and Wager, T. D. (2009). Modeling the hemodynamic response function in fMRI: Efficiency, bias and mis-modeling. *NeuroImage*, 45(1):S187–S198.
- Litman, L. and Davachi, L. (2008). Distributed learning enhances relational memory consolidation. *Learning & Memory*, 15(9):711–716.
- Little, R. J. A. and Rubin, D. B. (2019). *Statistical analysis with missing data*. Wiley series in probability and statistics. Wiley, Hoboken, NJ, third edition edition.
- Liu, R. and Koedinger, K. R. (2017). Towards Reliable and Valid Measurement of Individualized Student Parameters. *International Educational Data Mining Society*. Publisher: ERIC.
- Marr, D. (1971). Simple memory: a theory for archicortex. *Philosophical Transactions of the Royal Society of London. B, Biological Sciences*, 262(841):23–81.
- Martin, C. B., Douglas, D., Newsome, R. N., Man, L. L., and Barense, M. D. (2018). Integrative and distinctive coding of visual and conceptual object features in the ventral visual stream. *eLife*, 7:e31873.
- McKinney, W. (2010). Data Structures for Statistical Computing in Python. pages 56–61, Austin, Texas.
- Mishkin, M. (1978). Memory in monkeys severely impaired by combined but not by separate removal of amygdala and hippocampus. *Nature*, 273(5660):297–298.
- Mumford, J. A., Turner, B. O., Ashby, F. G., and Poldrack, R. A. (2012). Deconvolving BOLD activation in event-related designs for multivoxel pattern classification analyses. *NeuroImage*, 59(3):2636–2643.
- Munafò, M. R., Cremers, H. R., Wager, T. D., and Yarkoni, T. (2019). Power and design considerations in imaging research. In *Casting Light on the Dark Side of Brain Imaging*, pages 73–78. Elsevier.
- Murdock, B. B. (1962). The serial position effect of free recall. *Journal of Experimental Psychology*, 64(5):482–488.
- Myung, I. J. (2003). Tutorial on maximum likelihood estimation. *Journal of Mathematical Psychology*, 47(1):90–100.

- Nelson, T. O. and Dunlosky, J. (1991). When People's Judgments of Learning (JOLs) are Extremely Accurate at Predicting Subsequent Recall: The "Delayed-JOL Effect". *Psychological Science*, 2(4):267–271.
- Nelson, T. O. and Dunlosky, J. (1994). Norms of Paired-Associate Recall During Multi-trial Learning of Swahili-English Translation Equivalents. *Memory*, 2(3):325–335.
- Norman, K. A. and O'Reilly, R. C. (2003). Modeling hippocampal and neocortical contributions to recognition memory: A complementary-learning-systems approach. *Psychological Review*, 110(4):611–646.
- Nosofsky, R. M. (1988). Exemplar-based accounts of relations between classification, recognition, and typicality. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 14(4):700–708.
- O'Doherty, J. P., Dayan, P., Friston, K., Critchley, H., and Dolan, R. J. (2003). Temporal Difference Models and Reward-Related Learning in the Human Brain. *Neuron*, 38(2):329–337.
- Olkin, I. and Pratt, J. W. (1958). Unbiased Estimation of Certain Correlation Coefficients. *The Annals of Mathematical Statistics*, 29(1):201–211.
- Paller, K. A., Kutas, M., and Mayes, A. R. (1987). Neural correlates of encoding in an incidental learning paradigm. *Electroencephalography and Clinical Neurophysiology*, 67(4):360–371.
- Paller, K. A. and Wagner, A. D. (2002). Observing the transformation of experience into memory. *Trends in Cognitive Sciences*, 6(2):93–102.
- Palmeri, T. J., Love, B. C., and Turner, B. M. (2017). Model-based cognitive neuroscience. *Journal of Mathematical Psychology*, 76:59–64.
- Pavlik, P. I. and Anderson, J. R. (2008). Using a model to compute the optimal schedule of practice. *Journal of Experimental Psychology: Applied*, 14(2):101–117.
- Pereira, F. and Botvinick, M. (2011). Information mapping with pattern classifiers: A comparative study. *NeuroImage*, 56(2):476–496.
- Pitman, E. J. G. (1937). Significance Tests Which May be Applied to Samples From any Populations. *Supplement to the Journal of the Royal Statistical Society*, 4(1):119.

- Poldrack, R. A., Baker, C. I., Durnez, J., Gorgolewski, K. J., Matthews, P. M., Munafò, M. R., Nichols, T. E., Poline, J.-B., Vul, E., and Yarkoni, T. (2017). Scanning the horizon: towards transparent and reproducible neuroimaging research. *Nature Reviews Neuroscience*, 18(2):115–126.
- Power, J. D., Mitra, A., Laumann, T. O., Snyder, A. Z., Schlaggar, B. L., and Petersen, S. E. (2014). Methods to detect, characterize, and remove motion artifact in resting state fMRI. *NeuroImage*, 84:320–341.
- Rabiner, L. (1989). A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286.
- Ranganath, C., Johnson, M. K., and D’Esposito, M. (2003). Prefrontal activity associated with working memory and episodic long-term memory. *Neuropsychologia*, 41(3):378–389.
- Rau, M. A. and Pardos, Z. (2016). Adding eye-tracking AOI data to models of representation skills does not improve prediction accuracy. *International Educational Data Mining Society*, page 622. Publisher: ERIC.
- Reuter, M., Rosas, H. D., and Fischl, B. (2010). Highly accurate inverse consistent registration: A robust approach. *NeuroImage*, 53(4):1181–1196.
- Rissman, J., Greely, H. T., and Wagner, A. D. (2010). Detecting individual memories through the neural decoding of memory states and past experience. *Proceedings of the National Academy of Sciences*, 107(21):9849–9854.
- Rissman, J. and Wagner, A. D. (2012). Distributed Representations in Memory: Insights from Functional Brain Imaging. *Annual Review of Psychology*, 63(1):101–128.
- Ritchey, M., Montchal, M. E., Yonelinas, A. P., and Ranganath, C. (2015). Delay-dependent contributions of medial temporal lobe regions to episodic memory retrieval. *eLife*, 4:e05025.
- Rosenberg, M. D., Finn, E. S., Scheinost, D., Papademetris, X., Shen, X., Constable, R. T., and Chun, M. M. (2016). A neuromarker of sustained attention from whole-brain functional connectivity. *Nature Neuroscience*, 19(1):165–171.
- Rouder, J. N. and Lu, J. (2005). An introduction to Bayesian hierarchical models with an application in the theory of signal detection. *Psychonomic Bulletin & Review*, 12(4):573–604.

- Rubin, D. C. (1985). Memorability as a measure of processing: A unit analysis of prose and list learning. *Journal of Experimental Psychology: General*, 114(2):213–238.
- Rubin, D. C. and Friendly, M. (1986). Predicting which words get recalled: Measures of free recall, availability, goodness, emotionality, and pronunciability for 925 nouns. *Memory & Cognition*, 14(1):79–94.
- Ruch, T. C. (1928). Factors influencing the relative economy of massed and distributed practice in learning. *Psychological Review*, 35(1):19–45.
- Sanquist, T. F., Rohrbaugh, J. W., Syndulko, K., and Lindsley, D. B. (1980). Electrocor-tical Signs of Levels of Processing: Perceptual Analysis and Recognition Memory. *Psychophysiology*, 17(6):568–576.
- Schaefer, A., Kong, R., Gordon, E. M., Laumann, T. O., Zuo, X.-N., Holmes, A. J., Eickhoff, S. B., and Yeo, B. T. T. (2018). Local-Global Parcellation of the Human Cerebral Cortex from Intrinsic Functional Connectivity MRI. *Cerebral Cortex*, 28(9):3095–3114.
- Schmidt, F. L. (1971). The Relative Efficiency of Regression and Simple Unit Predictor Weights in Applied Differential Psychology. *Educational and Psychological Measurement*, 31(3):699–714.
- Scoville, W. B. and Milner, B. (1957). LOSS OF RECENT MEMORY AFTER BILAT-ERAL HIPPOCAMPAL LESIONS. *Journal of Neurology, Neurosurgery & Psychiatry*, 20(1):11–21.
- Stelzer, J., Chen, Y., and Turner, R. (2013). Statistical inference and multiple testing correction in classification-based multi-voxel pattern analysis (MVPA): Random permuta-tions and cluster size control. *NeuroImage*, 65:69–82.
- Stone, M. (1974). Cross-Validatory Choice and Assessment of Statistical Predictions. *Journal of the Royal Statistical Society: Series B (Methodological)*, 36(2):111–133.
- Tenison, C., Fincham, J. M., and Anderson, J. R. (2016). Phases of learning: How skill acquisition impacts cognitive processing. *Cognitive Psychology*, 87:1–28.
- Theobald, C. M. (1974). Generalizations of Mean Square Error Applied to Ridge Regres-sion. *Journal of the Royal Statistical Society: Series B (Methodological)*, 36(1):103–106.
- Treadway, M. T., Buckholtz, J. W., and Zald, D. H. (2013). Perceived stress predicts al-tered reward and loss feedback processing in medial prefrontal cortex. *Frontiers in Human Neuroscience*, 7.

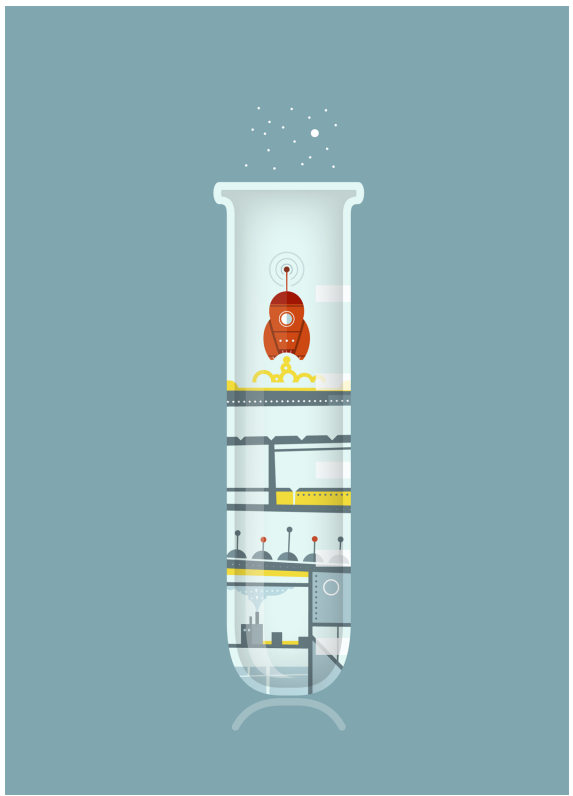
- Turner, B. M., Forstmann, B. U., and Steyvers, M. (2019). *Joint Models of Neural and Behavioral Data*. Computational Approaches to Cognition and Perception. Springer International Publishing, Cham.
- Turner, B. M., Forstmann, B. U., Wagenmakers, E.-J., Brown, S. D., Sederberg, P. B., and Steyvers, M. (2013). A Bayesian framework for simultaneously modeling neural and behavioral data. *NeuroImage*, 72:193–206.
- Turner, B. M., Rodriguez, C. A., Norcia, T. M., McClure, S. M., and Steyvers, M. (2016). Why more is better: Simultaneous modeling of EEG, fMRI, and behavioral data. *NeuroImage*, 128:96–115.
- Tustison, N. J., Avants, B. B., Cook, P. A., Yuanjie Zheng, Egan, A., Yushkevich, P. A., and Gee, J. C. (2010). N4ITK: Improved N3 Bias Correction. *IEEE Transactions on Medical Imaging*, 29(6):1310–1320.
- van De Sande, B. (2013). Properties of the Bayesian Knowledge Tracing Model. *Journal of Educational Data Mining*, 5(2):1–10. Publisher: ERIC.
- van der Linden, W. J. (2009). Conceptual Issues in Response-Time Modeling. *Journal of Educational Measurement*, 46(3):247–272.
- van der Walt, S., Colbert, S. C., and Varoquaux, G. (2011). The NumPy Array: A Structure for Efficient Numerical Computation. *Computing in Science & Engineering*, 13(2):22–30.
- van Maanen, L., Brown, S. D., Eichele, T., Wagenmakers, E.-J., Ho, T., Serences, J., and Forstmann, B. U. (2011). Neural Correlates of Trial-to-Trial Fluctuations in Response Caution. *Journal of Neuroscience*, 31(48):17488–17495.
- Vehtari, A., Gelman, A., and Gabry, J. (2017). Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. *Statistics and Computing*, 27(5):1413–1432.
- Visser, R. M., Scholte, H. S., Beemsterboer, T., and Kindt, M. (2013). Neural pattern similarity predicts long-term fear memory. *Nature Neuroscience*, 16(4):388–390.
- Wagner, A. D. (1998). Building Memories: Remembering and Forgetting of Verbal Experiences as Predicted by Brain Activity. *Science*, 281(5380):1188–1191.
- Wang, Q., Cagna, B., Chaminade, T., and Takerkart, S. (2020). Inter-subject pattern analysis: A straightforward and powerful scheme for group-level MVPA. *NeuroImage*, 204:116205.

- Wang, Y. and Heffernan, N. T. (2012). Leveraging First Response Time into the Knowledge Tracing Model. *International Educational Data Mining Society*. Publisher: ERIC.
- Ward, E. J., Chun, M. M., and Kuhl, B. A. (2013). Repetition Suppression and Multi-Voxel Pattern Similarity Differentially Track Implicit and Explicit Visual Memory. *Journal of Neuroscience*, 33(37):14749–14757.
- Weidemann, C. T. and Kahana, M. J. (2019). Neural measures of subsequent memory reflect endogenous variability in cognitive function. preprint, Neuroscience.
- Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L., François, R., Grolemond, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T., Miller, E., Bache, S., Müller, K., Ooms, J., Robinson, D., Seidel, D., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K., and Yutani, H. (2019). Welcome to the Tidyverse. *Journal of Open Source Software*, 4(43):1686.
- Wiecki, T. V., Sofer, I., and Frank, M. J. (2013). HDDM: Hierarchical Bayesian estimation of the Drift-Diffusion Model in Python. *Frontiers in Neuroinformatics*, 7.
- Wilson, R. C. and Niv, Y. (2015). Is Model Fitting Necessary for Model-Based fMRI? *PLOS Computational Biology*, 11(6):e1004237.
- Xiao, X., Dong, Q., Chen, C., and Xue, G. (2016). Neural pattern similarity underlies the mnemonic advantages for living words. *Cortex*, 79:99–111.
- Xie, W., Bainbridge, W. A., Inati, S. K., Baker, C. I., and Zaghoul, K. A. (2020). Memorability of words in arbitrary verbal associations modulates memory retrieval in the anterior temporal lobe. *Nature Human Behaviour*.
- Xue, G. (2018). The Neural Representations Underlying Human Episodic Memory. *Trends in Cognitive Sciences*, 22(6):544–561.
- Xue, G., Dong, Q., Chen, C., Lu, Z., Mumford, J. A., and Poldrack, R. A. (2010). Greater Neural Pattern Similarity Across Repetitions Is Associated with Better Memory. *Science*, 330(6000):97–101.
- Xue, G., Dong, Q., Chen, C., Lu, Z.-L., Mumford, J. A., and Poldrack, R. A. (2013). Complementary Role of Frontoparietal Activity and Cortical Pattern Similarity in Successful Episodic Memory Encoding. *Cerebral Cortex*, 23(7):1562–1571.
- Yarkoni, T. (2009). Big Correlations in Little Studies: Inflated fMRI Correlations Reflect Low Statistical Power—Commentary on Vul et al. (2009). *Perspectives on Psychological Science*, 4(3):294–298.

Yudelson, M. V., Koedinger, K. R., and Gordon, G. J. (2013). Individualized Bayesian Knowledge Tracing Models. In Lane, H. C., Yacef, K., Mostow, J., and Pavlik, P., editors, *Artificial Intelligence in Education*, pages 171–180, Berlin, Heidelberg. Springer Berlin Heidelberg.

Zhang, Q., Anderson, J. R., and Kass, R. E. (2016). Consistency in Brain Activation Predicts Success in Transfer. page 6.

Zhang, Y., Brady, M., and Smith, S. (2001). Segmentation of brain MR images through a hidden Markov random field model and the expectation-maximization algorithm. *IEEE Transactions on Medical Imaging*, 20(1):45–57.



THIS THESIS WAS TYPESET using L^AT_EX, originally developed by Leslie Lamport and based on Donald Knuth's T_EX. The body text is set in 11 point Egenolff-Berner Garamond, a revival of Claude Garamont's humanist typeface. The above illustration, "Science Experiment 02", was created by Ben Schlitter and released under CC BY-NC-ND 3.0. A template that can be used to format a PhD thesis with this look and feel has been released under the permissive MIT (X11) license, and can be found online at github.com/suchow/Dissertate or from its author, Jordan Suchow, at suchow@post.harvard.edu.